

On the Communication Complexity of PSM and CDS for Symmetric Functions

Reo Eriguchi¹

¹National Institute of Advanced Industrial Science and Technology (AIST), Japan
eriguchi-reo@aist.go.jp

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Abstract

Private Simultaneous Messages (PSM) and Conditional Disclosure of Secrets (CDS) are two fundamental primitives in information-theoretic cryptography. In a PSM protocol, each party sends a single message to a referee, who can then compute a function f of their private inputs but learns nothing else. A CDS protocol follows the same model as PSM, but the goal is to disclose a secret shared among all parties to the referee if and only if the function f evaluates to 1. Minimizing the communication complexity of these primitives is a central question in this area. Since the best known constructions for general functions require high communication complexity, recent studies have attempted to obtain more efficient constructions by focusing on symmetric functions, whose outputs are invariant under permutations of inputs. However, the extent to which exploiting symmetry can improve efficiency has remained unclear. In this work, we show upper and lower bounds that relate the optimal communication complexities of PSM and CDS for symmetric functions to those for general functions. When the number of parties is larger than the input domain size, our upper bound for PSM improves the best known communication complexity for symmetric functions. Furthermore, our upper bound for CDS demonstrates for the first time that symmetry can be exploited to reduce communication in CDS protocols. In contrast, when the number of parties is constant, our lower bounds show that focusing on symmetric functions yields only a constant-factor improvement. In addition, our new constructions for symmetric functions lead to improvements over the state-of-the-art results in related models such as ad hoc PSM and secret sharing.

1 Introduction

Private Simultaneous Messages (PSM) [26, 29] and *Conditional Disclosure of Secrets (CDS)* [28] are two fundamental primitives in information-theoretic cryptography. In a PSM protocol for a function f , each of n parties has an input x_i and shares common private randomness. Each party sends a single message to an external party called a referee, who can then compute $f(x_0, \dots, x_{n-1})$ but learns nothing else on the parties' inputs. In a CDS protocol, the parties also hold a common secret s . The goal is to disclose the secret s to the referee if and only if the function $f(x_0, \dots, x_{n-1})$ evaluates to 1. Note that in the CDS setting, the parties' inputs x_i are not private and are known to the referee. Despite their simplicity, PSM and CDS have been used as building blocks for many other cryptographic primitives, including constant-round secure computation [32, 31], attribute-based encryption [27], and secret sharing for general access structures [34, 5].

The central question in this area is to determine the optimal communication complexity of PSM and CDS protocols for a function f , where communication complexity is defined as the total bit length of parties' messages. This question has been extensively studied both for general functions and for functions with specific representations, e.g., [29, 4, 36, 17, 15, 7, 6, 8, 16]. Let $\mathcal{F}_{n,d}$ denote the set of general n -input functions where each input is taken from the domain of size d . Currently, the best known PSM protocol for $\mathcal{F}_{n,d}$ has communication complexity $O(d^{n/2}n^3)$ [15] and the best known CDS protocol has communication complexity $2^{\tilde{O}(\sqrt{n \log d})}$ [36]. However, these constructions are inefficient since they require at least sub-exponential communication in n , and are still far from optimal in comparison with the best known lower bounds [3, 8, 6].

A promising alternative research direction is to design efficient constructions by focusing on functions with specific structure. Among these, *symmetric functions*, whose outputs are invariant under permutations of the n inputs, are one of the most natural function classes with real-world applications such as elections and statistical analysis. We denote the subclass consisting of all symmetric functions by $\mathcal{S}_{n,d}$. For PSM protocols, several efficient constructions exploiting the symmetry of functions have been proposed [12, 18, 23, 40, 24, 37] and the best known communication complexity for $\mathcal{S}_{n,d}$ is $n^{2d/3+O(1)}$ [24]. This significantly outperforms the general constructions when the number n of parties is much larger than the domain size d (e.g., in an election, there are typically many more voters than candidates).

While symmetry can effectively reduce communication complexity, a natural question is to provide a *quantitative* analysis of the potential efficiency gain from focusing on symmetric functions, which is important for a clear understanding of their benefits. Specifically, we denote the optimal communication complexity of PSM and CDS for a function class $\mathcal{F}$ by $\text{PSM}(\mathcal{F})$ and $\text{CDS}(\mathcal{F})$, respectively. The goal of this work is to clarify the relation between the quantities $\mathfrak{C}(\mathcal{S}_{n,d})$ and $\mathfrak{C}(\mathcal{F}_{n,d})$, for $\mathfrak{C} \in \{\text{PSM}, \text{CDS}\}$. Existing works provide several direct upper bounds on $\text{PSM}(\mathcal{S}_{n,d})$, but the only known relation so far is the trivial inequality $\text{PSM}(\mathcal{S}_{n,d}) \leq \text{PSM}(\mathcal{F}_{n,d})$. Furthermore, to the best of our knowledge, no prior constructions of CDS have achieved any improvement in communication complexity by focusing on $\mathcal{S}_{n,d}$.

1.1 Our Results

In this work, we show upper and lower bounds that relate the optimal communication complexities of PSM and CDS protocols for symmetric functions to those for general functions. Consequently, we obtain close relations between the complexities of $\mathcal{S}_{n,d}$ and $\mathcal{F}_{n,d}$ in two extreme cases. When the number n of parties is constant, the communication complexity for $\mathcal{S}_{n,d}$ is equal to that for $\mathcal{F}_{n,d}$ up to a constant. This indicates a limitation on the potential efficiency gain achievable by focusing on symmetric functions. In contrast, when the domain size d is constant, the communication complexity for $\mathcal{S}_{n,d}$ is equal to that for $\mathcal{F}_{d-1,n+1}$ up to a factor of $O(n^2)$. This strengthens the previous results on the efficiency improvements of PSM for symmetric functions. In addition, our new constructions for symmetric functions lead to improvements over the state-of-the-art results in related models: (i) ad hoc PSM [11, 14], where only a subset of the parties actually send messages and (ii) a new efficient secret sharing scheme for multipartite uniform access structures See below for more precise statements.

Technically, our bounds are derived by two novel techniques. First, we propose a modular construction of PSM and CDS for symmetric functions. In the PSM case, this improves the communication complexity of the best known construction [24]. In the CDS case, this is the first result to exploit the symmetry of functions to reduce the communication complexity compared to general functions. Second, we introduce a generic technique of expanding the input domains of protocols, which may be of independent interest. This allows us to relate the optimal complexities for functions with different domain sizes. An overview of our techniques will be given in Section 2.

Relations to General Functions with Different Domains. Our first theorem relates the optimal communication complexities of PSM and CDS protocols for symmetric functions to those for general functions with different domain sizes.

Theorem 1.1. *For $\mathfrak{C} \in \{\text{PSM}, \text{CDS}\}$,*

$$\mathfrak{C}(\mathcal{F}_{n,d'}) \leq \mathfrak{C}(\mathcal{S}_{n,d}) \leq \mathfrak{C}(\mathcal{F}_{n,d}), \quad (1)$$

$$\mathfrak{C}(\mathcal{F}_{d-1,n'+1}) \leq \mathfrak{C}(\mathcal{S}_{n,d}) \leq O(n^2) \cdot \mathfrak{C}(\mathcal{F}_{d-1,n+1}), \quad (2)$$

where $d' = \lfloor d/n \rfloor$ and $n' = \lfloor n/(d-1) \rfloor$.

Proof idea. The upper bound in (1) is trivial as $\mathcal{S}_{n,d} \subseteq \mathcal{F}_{n,d}$. The upper bound in (2) is derived from our novel constructions of PSM and CDS protocols for symmetric functions. By definition, a symmetric function f can be expressed as a function of the *histogram* $\mathbf{h}_{\mathbf{x}}$ of inputs $\mathbf{x} = (x_0, \dots, x_{n-1})$, that is, $f(\mathbf{x}) = f^*(\mathbf{h}_{\mathbf{x}})$ for some f^* . We devise a modular method to reduce PSM for f to two components: PSM for f^* and basic PSM for message selection. Specifically, the parties first compute all possible messages for the PSM of f^* and then reveal only the messages needed to compute the output for $\mathbf{h}_{\mathbf{x}}$. The lower bounds follow from the

observation that symmetric functions are capable of computing general functions with smaller domains. This allows reducing PSM for general functions to symmetric ones.

By plugging in the best known upper bound for $\text{PSM}(\mathcal{F}_{n,d})$ [15], we obtain a new upper bound of $n^{d/2}$ for $\text{PSM}(\mathcal{S}_{n,d})$, ignoring a polynomial factor of n and d . This improves upon the state-of-the-art bound of $n^{2d/3}$ in [24]. For $\text{CDS}(\mathcal{S}_{n,d})$, we obtain an upper bound of $n^2 2^{\tilde{O}(\sqrt{d} \log n)}$. This improves upon the previously known bound of $2^{\tilde{O}(\sqrt{n \log d})}$, derived from the best upper bound for $\mathcal{F}_{n,d}$ [36], when $n \gg d$. To the best of our knowledge, this is the first result to exploit the symmetry of functions in reducing the communication complexity of CDS protocols.

On the other hand, our lower bound in (1) indicates a limitation of focusing on symmetric functions: the communication complexity of CDS and PSM for $\mathcal{S}_{n,d}$ cannot be smaller than that for general functions with inputs shorter by $O(\log n)$ bits. An implication of (2) is that any further improvement of our new upper bound for symmetric functions requires improving the best known bounds for general functions [36, 15]. For example, an upper bound of n^{cd} on $\text{PSM}(\mathcal{S}_{n,d})$ with $c < 1/2$ would imply an upper bound of $O_d(n^{cd})$ on $\text{PSM}(\mathcal{F}_{d-1,n+1})$. Therefore, improving the exponent in our upper bound requires improving the exponent of the dominant term $d^{n/2}$ in the best known bound for $\mathcal{F}_{n,d}$ [15].

Towards Tight Relations to General Functions. Our next theorem relates the optimal communication complexity of $\mathcal{S}_{n,d}$ to quantities solely based on $\mathcal{F}_{n,d}$ or $\mathcal{F}_{d-1,n+1}$.

Theorem 1.2. For $\mathfrak{C} \in \{\text{PSM}, \text{CDS}\}$,

$$\alpha_n \cdot \mathfrak{C}(\mathcal{F}_{n,d}) \leq \mathfrak{C}(\mathcal{S}_{n,d}) \leq \mathfrak{C}(\mathcal{F}_{n,d}), \quad (3)$$

$$\beta_d \cdot \mathfrak{C}(\mathcal{F}_{d-1,n+1}) \leq \mathfrak{C}(\mathcal{S}_{n,d}) \leq O(n^2) \cdot \mathfrak{C}(\mathcal{F}_{d-1,n+1}), \quad (4)$$

where $\alpha_n = n^{-O(n)}$ and $\beta_d = d^{-O(d)}$.

As a consequence, we obtain a close relation between the optimal complexities of PSM and CDS for symmetric and general functions in the two extreme cases: for $\mathfrak{C} \in \{\text{PSM}, \text{CDS}\}$,

- If $n = O(1)$, $\mathfrak{C}(\mathcal{S}_{n,d})$ is equal to $\mathfrak{C}(\mathcal{F}_{n,d})$ up to a constant.
- If $d = O(1)$, $\mathfrak{C}(\mathcal{S}_{n,d})$ is equal to $\mathfrak{C}(\mathcal{F}_{d-1,n+1})$ up to a factor of $O(n^2)$.

An important implication of (3) is that, for PSM and CDS protocols with a constant number of parties, the efficiency gain achievable by focusing on symmetric functions is limited to a constant-factor improvement. This can also be explained intuitively. Suppose, heuristically, that the worst-case complexity of PSM and CDS is determined by the maximum representation size of functions, as suggested by existing results. The worst-case representation size for $\mathcal{F}_{n,d}$ is d^n , whereas it is $\binom{n+d-1}{d-1}$ for $\mathcal{S}_{n,d}$. Since $\binom{n+d-1}{d-1} \geq d^n/n^n$ when $n = O(1)$, the complexity for symmetric functions can be smaller than that for general functions only by a constant factor.

Proof idea. We show a relation between the complexities of PSM or CDS for functions with different domain sizes. Specifically, for $d' < d$, we prove an inequality of the form $\mathfrak{C}(\mathcal{F}_{n,d}) \leq M \cdot \mathfrak{C}(\mathcal{F}_{n,d'})$ for some M depending only on n . To this end, we devise a novel method to construct a PSM (resp. CDS) protocol for $\mathcal{F}_{n,d}$ using only PSM (resp. CDS) protocols for $\mathcal{F}_{n,d'}$; that is, a technique of expanding the input domains. We can then convert the lower bounds in Theorem 1.1 to those solely based on $\mathcal{F}_{n,d}$ or $\mathcal{F}_{d-1,n+1}$.

Applications to Related Models. Our new PSM and CDS protocols for symmetric functions have respective applications to ad hoc PSM [11, 14] and secret sharing [39, 33]. First, ad hoc PSM is an advanced notion of PSM, in which only a subset S of k parties actually send messages, and the referee evaluates a k -input function f on that subset of inputs. It is natural in this model to assume that f is a symmetric function independent of the identities of the parties. There exist generic compilers from PSM to ad hoc PSM [2, 19], which incur only a polynomial overhead in k and n to the complexity of the k -party PSM for f . However, the resulting bounds incurred an exponential factor of $d^{k/2}$ in k , since they relied on the trivial bound obtained from $\text{PSM}(\mathcal{F}_{k,d})$. Our improved bound on $\text{PSM}(\mathcal{S}_{k,d})$ immediately implies an ad hoc PSM protocol with communication complexity dominated by $k^{d/2}$, which is polynomial in k when $d = O(1)$. In addition to this immediate application, we also propose a direct construction of ad hoc PSM for symmetric functions, without relying on the generic compiler of [2]. As a result, we further improve the upper bound by a factor of k .

Second, it was shown in [2] that k -party CDS protocols directly imply secret sharing schemes for uniform access structures, in which all sets of size larger than k can recover a secret and all sets of size smaller than k cannot. We show that our CDS protocol for symmetric functions yields an improvement in the share size for a special class of *multipartite* uniform access structures. An access structure is called d -partite [25] if the set of parties is partitioned into d parts and parties within the same part have equivalent roles in the recoverability of secrets¹. We obtain a secret sharing scheme for d -partite k -uniform access structures whose share size is asymptotically smaller than $k^{\tilde{O}(\sqrt{d})}$. As a comparison, the general compiler of [2] results in the share size incurring a sub-exponential factor $2^{\tilde{O}(\sqrt{k \log n})}$ in k ; and a folklore construction for arbitrary d -partite access structures based on [33] results in share size $O(k^{d-1})$.

1.2 Related Works

We briefly introduce previous works on the communication complexity of PSM and CDS protocols. Feige, Kilian, and Naor [26] formalized the notion of PSM and presented general feasibility results in the two-party setting, and Ishai and Kushilevitz [29] extended it to the multiparty setting. There exist many constructions of PSM for general functions and for those represented in certain computational models such as branching programs [29] and arithmetic formulas [4] (see [30] for a survey). Beimel et al. [13] successfully devised a more efficient two-party PSM protocol for general functions and this was later generalized and improved by [15, 7], leading to the currently best known complexity. To mitigate the exponential communication, PSM protocols tailored to symmetric functions have also been proposed [12, 18, 23, 40, 24, 37]. On the lower-bound side, Applebaum et al. [3] proved the first non-trivial lower bound on the communication complexity of PSM and Ball and Randolph [8] extended it to the multiparty setting. The notion of CDS was originally introduced by Gertner et al. [28]. Since then, many works have aimed at improving communication complexity, e.g., [35, 36, 17, 16]. Applebaum and Vasudevan [6] showed interesting connections between two-party CDS and other communication games, leading to non-trivial lower bounds. In [17] and [10], nearly tight upper and lower bounds were shown for a restricted class of linear CDS. Despite all these progresses, however, closing the large gaps between the upper and lower bounds in the general setting remains a long-standing open question.

2 Technical Overview

2.1 Proof of Theorem 1.1

Upper Bounds. The right inequality of (2) is derived by our novel efficient constructions of PSM and CDS protocols for symmetric functions. First, we consider the case of PSM. All the previous constructions [12, 23, 24] exploited the symmetry of f to obtain its compact representation in certain computation models (e.g., branching programs or truth-tables), and then reduced to the best known PSM protocols in the respective models. However, such “native” approaches can derive only a direct upper bound on $\text{PSM}(\mathcal{S}_{n,d})$ and cannot yield any non-trivial relation between the optimal communication complexities of $\mathcal{S}_{n,d}$ and other classes. In contrast, we show a generic construction of a PSM protocol for $\mathcal{S}_{n,d}$ from one for $\mathcal{F}_{d-1,n+1}$.

If f is symmetric, then $f(\mathbf{x})$ is uniquely determined by the *histogram* $\mathbf{h}_{\mathbf{x}} = (h^{(0)}, \dots, h^{(d-1)})$ of $\mathbf{x} = (x_0, \dots, x_{n-1})$, where $h^{(j)}$ counts the frequency of a value j in $\mathbf{x}$. Thus, there exists a function $f^* : [n+1]^d \rightarrow \{0, 1\}$ such that

$$f^*(\mathbf{h}_{\mathbf{x}}) = f(\mathbf{x}), \quad \forall \mathbf{x} \in [d]^n. \quad (5)$$

Since $h^{(d-1)} = n - \sum_{j=0}^{d-2} h^{(j)}$, only the $d-1$ variables $(h^{(j)})_{j \in [d-1]}$ are free and we may thus assume that $f^* \in \mathcal{F}_{d-1,n+1}$.

Suppose that a PSM protocol Π_{f^*} for f^* is given, from which we construct a PSM protocol Π_f for f . Let $P_0, \dots, P_{n-1}$ denote the n real parties involved in Π_f and $Q^{(0)}, \dots, Q^{(d-2)}$ denote the $d-1$ virtual parties in Π_{f^*} . Suppose that the real parties receive private randomness r for Π_{f^*} in advance. Each P_i computes the messages $m^{(j)}(h)$ that a virtual party $Q^{(j)}$ in Π_{f^*} computes on all possible inputs $h \in [n+1]^d$ and randomness

¹Note that this should not be confused with another definition of “multipartite” access structures in the literature [1, 10], where the set is partitioned into k parts and every authorized set of size k contains exactly one party from each parts.

r . Let $(h^{(0)}, \dots, h^{(d-2)}) = \mathbf{h}_x$ denote the histogram of real parties' inputs $\mathbf{x}$. If the parties are able to reveal the message $m^{(j)}(h^{(j)})$ (and nothing else) to the referee, then he can recover $f^*(\mathbf{h}_x) = f(\mathbf{x})$ using the decoding algorithm of Π_{f^*} , which completes the protocol Π_f . A key observation is that each bin $h^{(j)}$ of the histogram is expressed as $h^{(j)} = \sum_{i \in [n]} \delta_j(x_i)$, where $\delta_j(x) = 1$ if $x = j$ and $\delta_j(x) = 0$ otherwise. Furthermore, $\delta_j(x_i)$ can be locally computed by P_i . Thus, the task of revealing $m^{(j)}(h^{(j)})$ for each j is essentially reduced to the following basic functionality, which we call the *selection function*:

$$\text{Sel}_n(\mathbf{m}; a_0, \dots, a_{n-1}) := \mathbf{m}[\sum_{i \in [n]} a_i],$$

where $\mathbf{m} = (\mathbf{m}[0], \dots, \mathbf{m}[n]) \in \{0, 1\}^{n+1}$ is commonly held by all parties and $a_i \in \{0, 1\}$ is held by P_i . This functionality was originally introduced by [18] in a different context, and efficient constructions achieving communication complexity $O(n^2)$ are known [18, 40]. Since these protocols can also be applied in our purpose of retrieving an entry corresponding to the sum of parties' input bits, we obtain the inequality $\text{PSM}(f) \leq O(n^2) \cdot \text{PSM}(f^*)$. We note that the protocol in [40] achieves the currently best communication complexity $O(n^2)$ but is not computationally efficient. Nevertheless, this issue can be mitigated by using the protocol in [18], which achieves polynomial computational complexity at the cost of slightly increasing communication complexity.

Similarly, we can construct a CDS protocol for $f \in \mathcal{S}_{n,d}$ from an arbitrary CDS protocol for the associated function $f^* \in \mathcal{F}_{d-1,n+1}$. A difference is that when evaluating the selection function Sel_n , it is unnecessary to hide the input bits $a_0, \dots, a_{n-1}$ in CDS protocols, whereas the messages $\mathbf{m}$ are still private. We introduce this weaker functionality as the selection function with public inputs, denoted by pSel_n . By leveraging this relaxed setting, we present a PSM protocol for pSel_n whose communication and computation complexities are both $O(n^2)$. This should be contrasted with the best known PSM protocol for Sel_n with $O(n^2)$ communication [40], whose computational complexity is $2^{O(n)}$.

Lower Bounds. To derive the left inequality in (1), it suffices to show that for any function $g \in \mathcal{F}_{n,d'}$, there exists a symmetric function $f \in \mathcal{S}_{n,d}$ such that $\text{PSM}(g) \leq \text{PSM}(f)$, that is, any protocol for f can be transformed to a protocol for g with the same communication complexity. Since the following argument can apply to the CDS setting without modification, we only consider the PSM case here. Our approach is to find appropriate embeddings $\psi_i : [d'] \rightarrow [d]$ and set f as any function such that $f(\psi_0(x_0), \dots, \psi_{n-1}(x_{n-1})) = g(x_0, \dots, x_{n-1})$. Then, any PSM protocol for f can be used to compute g : each party with input $x_i \in [d']$ runs the protocol for f with input $\psi_i(x_i) \in [d]$. Note that the naive embedding $\psi_i(x) = x$ does not satisfy the requirement since g is not necessarily symmetric. Instead, we embed the i^{th} input $x_i \in [d']$ into an element of $[d]$, storing the index i in an additional input slot of size $d/d' \geq n$. For the ease of exposition, we assume that n divides d . We can decompose the input domain of each party as $[d] = [n] \times [d'] = \{(i, x) : i \in [n], x \in [d']\}$ and define $\psi_i(x) = (i, x)$ for $i \in [n], x \in [d']$. Then, we set f as a function that first sorts inputs $(i_0, x_0), \dots, (i_{n-1}, x_{n-1})$ by the indices i_j 's and then outputs $g(x_{j_0}, \dots, x_{j_{n-1}})$, where $((i_0, x_{j_0}), \dots, (i_{n-1}, x_{j_{n-1}}))$ is the sorted inputs. (We set a value of f to 0 if inputs contain identical indices.) By definition, f satisfies the requirement that $f(\psi_0(x_0), \dots, \psi_{n-1}(x_{n-1})) = g(x_0, \dots, x_{n-1})$ and is symmetric as it sorts inputs at the beginning.

Similarly, the proof of the left inequality in (2) is reduced to showing that for any function $g \in \mathcal{F}_{d-1,n'+1}$, there exists $f \in \mathcal{S}_{n,d}$ such that any PSM protocol for f can be transformed to a protocol for g preserving communication complexity. Note that an input to g can be viewed as a histogram $\mathbf{h} = (h^0, \dots, h^{(d-2)})$ with $d-1$ bins. We have chosen $n' = \lfloor n/(d-1) \rfloor$ to ensure that the sum $\sum_{j \in [d-1]} h^{(j)}$ does not exceed n for any input $\mathbf{h} \in [n'+1]^{d-1}$. Therefore, we can find a symmetric function $f \in \mathcal{S}_{n,d}$ whose associated function f^* "contains" g , that is, $f^*(\mathbf{h}) = g(\mathbf{h})$ for every $\mathbf{h} \in [n'+1]^{d-1}$. We construct a PSM protocol Π_g for g from any protocol Π_f for f . For ease of exposition, we assume that $d-1$ divides n . The construction proceeds in the *reverse* direction of our previous construction for deriving the upper bound. Let $\{Q^{(j)} : j \in [d-1]\}$ denote the $d-1$ real parties involved in Π_g and $\{P_i^{(j)} : i \in [n'], j \in [d-1]\}$ denote the n virtual parties in Π_f . Each party $Q^{(j)}$ simulates the roles of $\{P_i^{(j)} : i \in [n']\}$. He encodes his input $h^{(j)} \in [n'+1]$ into a vector $(x_0^{(j)}, \dots, x_{n'-1}^{(j)}) = (\underbrace{j, \dots, j}_{h^{(j)}}, \underbrace{d-1, \dots, d-1}_{n'-h^{(j)}})$, and then computes a message that $P_i^{(j)}$ computes on input $x_i^{(j)}$. The referee receives the messages encoding the inputs $\mathbf{x} = (x_i^{(j)})_{i \in [n'], j \in [d-1]}$ where a value $j \in [d-1]$

appears $h^{(j)}$ times and $d - 1$ appears $\sum_j (n' - h^{(j)}) = n - \sum_j h^{(j)}$ times. Since the histogram of $\mathbf{x}$ is equal to $\mathbf{h}$, the decoding of the messages gives $f(\mathbf{x}) = f^*(\mathbf{h}) = g(\mathbf{h})$, which completes the protocol Π_g .

2.2 Proof of Theorem 1.2

The Case of CDS. To lower bound $\text{CDS}(\mathcal{F}_{n,d'})$ by $\text{CDS}(\mathcal{F}_{n,d})$ for some $d > d'$, we present a technique for relating the optimal communication complexities of CDS with different domain sizes. Specifically, we prove an inequality of the form $\text{CDS}(\mathcal{F}_{n,d_2}) \leq M \cdot \text{CDS}(\mathcal{F}_{n,d_1})$ for $d_2 > d_1$, where M is a constant depending solely on n . Such bounds have not been considered so far in either the CDS or PSM setting, as the previous works only provided direct upper bounds for $\mathcal{F}_{n,d}$. To this end, we show a construction of a CDS protocol for any function $F \in \mathcal{F}_{n,d_2}$ based on CDS protocols for functions in $\mathcal{F}_{n,d_1}$; that is, a technique of expanding input domains. A key difficulty is that *we cannot assume any structure on given protocols for $\mathcal{F}_{n,d_1}$* . This is because the structure of protocols achieving the optimal communication complexity $\text{CDS}(\mathcal{F}_{n,d_1})$ is currently totally unclear. Therefore, we need to devise a generic construction applicable to arbitrary CDS protocols for $\mathcal{F}_{n,d_1}$.

Here, we show a core construction of expanding each input domain by one bit. Let $F \in \mathcal{F}_{n,2d}$. We decompose the input domain $[2d]$ as $[d] \times \{0, 1\}$. Then, F can be expressed as

$$F((x_i, y_i)_{i \in [n]}) = \sum_{\mathbf{w} \in \{0,1\}^n} f_{\mathbf{w}}(\mathbf{x}) \cdot \delta_{w_0}(y_0) \cdots \delta_{w_{n-1}}(y_{n-1}) \quad (6)$$

for any $\mathbf{x} \in [d]^n, \mathbf{y} \in \{0, 1\}^n$, where $f_{\mathbf{w}}$ is the function $F(\cdot, \mathbf{w})$ obtained by fixing each y -component to w_i . Note that $f_{\mathbf{w}} \in \mathcal{F}_{n,d}$ and each $\delta_{w_i}(y_i)$ can be locally computed by party i holding y_i . Observe that $F(\mathbf{x}, \mathbf{y}) = 1$ if and only if there uniquely exists $\mathbf{w}$ such that $f_{\mathbf{w}}(\mathbf{x})\delta_{\mathbf{w}}(\mathbf{y}) = 1$. A CDS protocol for F is thus obtained by executing 2^n CDS protocols, each for the function $f_{\mathbf{w}}(\mathbf{x})\delta_{\mathbf{w}}(\mathbf{y})$. In the sub-protocol corresponding to $\mathbf{w} = \mathbf{y}$, the referee learns the secret if and only if $F(\mathbf{x}, \mathbf{y}) = 1$. In all other sub-protocols, the referee learns nothing since $f_{\mathbf{w}}(\mathbf{x})\delta_{\mathbf{w}}(\mathbf{y}) = 0$ necessarily holds.

Thus, it suffices to construct a CDS protocol for

$$G_f(\mathbf{x}, \mathbf{y}) := f(\mathbf{x}) \cdot y_0 \cdots y_{n-1}$$

from a CDS protocol Π_f for f . Suppose that each party with input (x_i, y_i) computes a message that encodes an input x_i and a secret s , using Π_f . If $y_i = 1$ for all i and the parties agree on it, then the predicate G_f is equivalent to f and hence the security is guaranteed by Π_f . However, the parties cannot determine in advance whether $y_i = 1$ for all i , and if $y_i = 0$ for some i , then G_f always evaluates to 0 and the secret s must be hidden. Our solution is simply to have party i send nothing if $y_i = 0$. In this case, the referee misses at least one message and cannot proceed to the decoding phase of Π_f . Although a subset of messages may leak information about the secret in plain CDS protocols, it is known that plain CDS can be efficiently upgraded into *strong* CDS [21], in which no information about secrets is leaked as long as at least one of the messages is missing. Therefore, if we use a strong CDS protocol for f as Π_f , then the referee cannot learn the secret whenever $\prod_{i \in [n]} y_i = 0$ (even if $f(\mathbf{x}) = 1$). Since the transformation in [21] incurs a communication overhead of n , we obtain $\text{CDS}(F) \leq n2^n \cdot \max_{f \in \mathcal{F}_{n,d}} \text{CDS}(f)$, yielding an inequality $\text{CDS}(\mathcal{F}_{n,2d}) \leq n2^n \cdot \text{CDS}(\mathcal{F}_{n,d})$.

The Case of PSM. Next, we relate the optimal communication complexities of PSM with different domain sizes. We aim at constructing a PSM protocol for any $F \in \mathcal{F}_{n,d_2}$ based on PSM protocols for functions in $\mathcal{F}_{n,d_1}$. We here focus on expanding the domain by one bit, that is, $d_2 = 2d$ and $d_1 = d$. Let $F \in \mathcal{F}_{n,2d}$. We attempt to exploit the equality (6) again and reduce PSM for F to $f \in \mathcal{F}_{n,d}$ in a manner similar to the CDS case. However, this does not work for PSM protocols since the referee can immediately learn the inputs $\mathbf{y}$ from the index $\mathbf{w}$ corresponding to the sub-protocol in which the referee receives all messages. Moreover, the issue cannot be addressed even if the 2^n sub-protocols are randomly permuted. This is because the decoding algorithm of a PSM protocol may depend on the description of the function being computed (i.e., $f_{\mathbf{w}}$ in this case), and hence it is impossible to make the referee fully oblivious to the index $\mathbf{w}$.

Alternatively, we consider a PSM protocols for computing each summand $f_{\mathbf{w}}(\mathbf{x})\delta_{w_0}(y_0) \cdots \delta_{w_{n-1}}(y_{n-1}) + b_{\mathbf{w}}$. Here, the $b_{\mathbf{w}}$'s are random additive shares of zero, which are included in common randomness. The addition by $b_{\mathbf{w}}$ ensures that the referee learns the total sum only. Therefore, the computation of F is reduced

to 2^n independent invocations of a PSM protocol for a function G_f^{plus} defined as

$$G_f^{\text{plus}}(b; \mathbf{x}, \mathbf{y}) := f(\mathbf{x}) \cdot y_0 \cdots y_{n-1} + b,$$

where $f \in \mathcal{F}_{n,d}$ and b is a bit known to all parties. We thus obtain $\text{PSM}(\mathcal{F}_{n,2d}) \leq 2^n \cdot \max_{f \in \mathcal{F}_{n,d}} \text{PSM}(G_f^{\text{plus}})$.

We reduce a PSM protocol for G_f^{plus} to PSM protocols for functions in $\mathcal{F}_{n,d}$. Our key observation is that G_f^{plus} can be expressed as

$$G_f^{\text{plus}}(b; \mathbf{x}, \mathbf{y}) = H(b, f(\mathbf{x}), \mathbf{y})$$

using a function $H(b, z, \mathbf{y}) := zy_0 \cdots y_{n-1} + b$. Intuitively, this decomposes the computation of G_f^{plus} into two parts: the computation of f and that of H , whose complexity depends only on n . It is then sufficient to let the referee learn the value of $H(b, f(\mathbf{x}), \mathbf{y})$ and nothing else. For that, we use a *decomposable randomized encoding* (DRE) of H , which is an information-theoretic variant of garbling schemes [30]. Specifically, based on private randomness ρ , it generates a garbled function $\hat{H}$ encoding H and garbled inputs $\hat{b}, \hat{z}, \hat{\mathbf{y}}$ encoding $b, z, \mathbf{y}$ (respectively) such that $\hat{H}(\hat{b}, \hat{z}, \hat{\mathbf{y}}) = H(b, z, \mathbf{y})$ but they leak nothing else. Since the complexity of H is independent of d , the size of garbled functions and inputs is also independent of d (and is actually polynomial in n) [30, 20]. It thus suffices to convey $\hat{H}$, $\hat{b}$, $\hat{z}$, and $\hat{\mathbf{y}}$ to the referee. It is easy to reveal $\hat{H}$, $\hat{b}$ and each $\hat{y}_i$ since a party who has b, y_i , and ρ can locally compute them on his own. The remaining task is to reveal a garbled input $\hat{z}$ corresponding to $z = f(\mathbf{x})$, which depends on $\mathbf{x}$ jointly held by all parties. We leverage an additional property that the DRE can be made *affine*, ensuring that $\hat{z}$ takes the form $\alpha f(\mathbf{x}) + \beta$, where α and β are functions of ρ only. The task is then completed by invoking a PSM protocol for the function that augments f as

$$f^{\text{affine}}(a, b; x_0, \dots, x_{n-1}) = af(x_0, \dots, x_{n-1}) + b,$$

where (a, b) are two auxiliary bits commonly known to all parties but hidden from the referee. We can thus obtain $\text{PSM}(G_f^{\text{plus}}) \leq \text{PSM}(f^{\text{affine}}) \cdot \text{poly}(n)$, where the factor $\text{poly}(n)$ comes from the size of the DRE $\hat{H}$.

Note that f^{affine} can be viewed as a composition of f with a secret affine function $A(X) = aX + b$, representing a richer functionality than f itself. In particular, a PSM protocol for f^{affine} needs to keep two additional bits a and b secret, compared to the case of computing the original f . We show that the number of additional secret bits can be reduced by one. Specifically, a PSM protocol for f^{affine} can be reduced to one for the function

$$f^{\text{plus}}(b; x_0, \dots, x_{n-1}) = f(x_0, \dots, x_{n-1}) + b,$$

where a bit b is commonly known to all parties except for the referee. Assume for simplicity that f is not the constant function that always outputs 1, and fix an input $\mathbf{x}^* = (x_i^*)_{i \in [n]}$ such that $f(\mathbf{x}^*) = 0$. Observe that $f^{\text{affine}}(1, b; \mathbf{x}) = f^{\text{plus}}(b; \mathbf{x})$ and $f^{\text{affine}}(0, b; \mathbf{x}) = b = f^{\text{plus}}(b; \mathbf{x}^*)$. Since the value of a is known to all parties, they can, without coordination, agree to input $\mathbf{x}$ to a PSM protocol for f^{plus} when $a = 1$, and $\mathbf{x}^*$ when $a = 0$. This realizes a PSM protocol for f^{affine} and hence yields $\text{PSM}(f^{\text{affine}}) \leq \text{PSM}(f^{\text{plus}})$.

What remains is to upper bound the ratio between $\text{PSM}(\mathcal{F}_{n,d}^{\text{plus}})$ and $\text{PSM}(\mathcal{F}_{n,d})$ by a constant depending only on n . Heuristically, this ratio is expected to be small when the computation model is restricted: a formula computing f^{plus} requires only one additional gate compared to f , and the size of a truth table representing f^{plus} is only twice as large as that of f . Since the existing PSM protocols have polynomial communication complexity in the function representation size in the respective models [4, 15], the multiplicative gap between the communication complexities for f^{plus} and f remains constant. However, as we said earlier, we cannot assume any structures of PSM protocols to derive a relation between the optimal complexities $\text{PSM}(\mathcal{F}_{n,d}^{\text{plus}})$ and $\text{PSM}(\mathcal{F}_{n,d})$. A formal proof would thus require a general technique that does not rely on any specific computational model. A PSM protocol for f^{plus} can be viewed as revealing either $f(\mathbf{x})$ or its negation $\bar{f}(\mathbf{x})$ to the referee, without telling which one is being revealed. A naive approach would be to run the protocols for f and $\bar{f}$ in parallel but in a random order, which might seem to hide the value of b from the referee. However, since the decoding algorithm of a PSM protocol may depend on the description of the function being computed, the referee may learn b even if the order of protocol executions is randomly permuted.

We show that a PSM protocol for f^{plus} with $f \in \mathcal{F}_{n,d}$ can be constructed based on PSM protocols for functions in $\mathcal{F}_{n,d}$ with a constant overhead. This implies that the ratio between $\text{PSM}(\mathcal{F}_{n,d}^{\text{plus}})$ and $\text{PSM}(\mathcal{F}_{n,d})$

is at most a constant. First, a PSM protocol for f^{plus} can be directly obtained from a PSM protocol for a function $g : [2d] \times [d]^{n-1} \rightarrow \{0, 1\}$ defined as

$$g((x_0, b), x_1, \dots, x_{n-1}) = f^{\text{plus}}(b; x_0, x_1, \dots, x_{n-1})$$

for any $x_i \in [d]$ and $b \in \{0, 1\}$. Indeed, party 0 just inputs $(x_0, b) \in [d] \times \{0, 1\} = [2d]$ to the protocol for g .

It thus suffices to consider a function g with domain $[2d] \times [d]^{n-1}$. We decompose the domain of party 0 as $[2d] = [2d/c] \times [c]$ for a constant c specified later (assuming that $2d$ is a multiple of c). That is, any element $x_0 \in [2d]$ is uniquely represented as (x'_0, y) where $x'_0 \in [2d/c]$ and $y \in [c]$. Similarly to Eq. (6), we have

$$g((x'_0, y), x_1, \dots, x_{n-1}) = \sum_{w \in [c]} g_w(x'_0, x_1, \dots, x_{n-1}) \cdot \delta_w(y) \quad (7)$$

for any $(x'_0, y) \in [2d/c] \times [c]$ and $x_i \in [d]$, where $g_w(x'_0, x_1, \dots, x_{n-1}) = g((x'_0, w), x_1, \dots, x_{n-1})$. Furthermore, setting $K((z_w)_{w \in [c]}, (\delta_w)_{w \in [c]}) = \sum_{w \in [c]} z_w \delta_w$, we can express Eq. (7) as

$$g((x'_0, y), x_1, \dots, x_{n-1}) = K((g_w(x'_0, x_1, \dots, x_{n-1}))_{w \in [c]}, (\delta_w(y))_{w \in [c]}).$$

In a similar manner to the reduction from G_f^{plus} to f^{affine} , a PSM protocol for g is obtained from PSM protocols for the compositions g_w^{affine} of g_w with affine functions. Note that if $c = O(1)$, then K is a function of constant size, which implies that it admits a decomposable affine randomized encoding of constant size. Hence, the overall overhead is constant. Since a PSM protocol for g_w^{affine} is further reduced to one for g_w^{plus} without any overhead, we have shown that a PSM protocol for g can be obtained from PSM protocols for the functions g_w^{plus} . Observe that g_w^{plus} has a smaller domain, taking inputs $(b; x'_0, x_1, \dots, x_{n-1})$ from $\{0, 1\} \times [2d/c] \times [d]^{n-1}$, where b is a common input. By including the bit b in party 0's input, a PSM protocol for g_w^{plus} is directly derived from a protocol for some function with domain $[4d/c] \times [d]^{n-1}$. Therefore, by setting $c = 4$, a PSM protocol for g is reduced to a PSM protocol for functions in $\mathcal{F}_{n,d}$, and the overhead depends only on the constant c .

In summary, PSM for any $F \in \mathcal{F}_{n,2d}$ have been reduced to PSM for functions in $\mathcal{F}_{n,d}$ as follows:

$$F \in \mathcal{F}_{n,2d} \xrightarrow{2^n \times} G_f^{\text{plus}} \ (f \in \mathcal{F}_{n,d}) \xrightarrow{\text{poly}(n) \times} \mathcal{F}_{n,d}^{\text{affine}} \xrightarrow{1 \times} \mathcal{F}_{n,d}^{\text{plus}} \xrightarrow{O(1) \times} \mathcal{F}_{n,d},$$

where $\xrightarrow{m \times}$ indicates that the reduction increases communication complexity by m . We thus obtain an inequality $\text{PSM}(\mathcal{F}_{n,2d}) \leq O(2^n \cdot \text{poly}(n)) \cdot \text{PSM}(\mathcal{F}_{n,d})$.

3 Preliminaries

We define $[n] = \{0, 1, \dots, n-1\}$ for $n \in \mathbb{N}$ as it will be more convenient to start indexing at 0. For a set X , we write $x \leftarrow_{\$} X$ if x is sampled uniformly at random from X . Let $\mathbb{F}_2 = \{0, 1\}$ denote the finite field of size 2. For $n \in \mathbb{N}$, let $\mathbb{Z}_n$ denote the ring of integers modulo n . For a function $f(n)$, we write $O_t(f(n))$ to denote the set of functions that are asymptotically bounded above by $f(n)$ up to a constant depending on a parameter t but not on n . The notation $\tilde{O}(f(n))$ hides polylogarithmic factors in $f(n)$. We denote the bit length of a string S by $|S|$. Throughout the paper, we assume by default that n denotes the number of parties and d denotes the cardinality of each party's input domain.

3.1 Symmetric Functions

For positive integers $d_0, \dots, d_{n-1}$, we define $\mathcal{F}_{(d_0, \dots, d_{n-1})}$ as the set of all functions $f : [d_0] \times \dots \times [d_{n-1}] \rightarrow \{0, 1\}$. If $d_0 = \dots = d_{n-1} = d$, then we simply write $\mathcal{F}_{n,d} = \mathcal{F}_{(d, \dots, d)}$. We call a function $f \in \mathcal{F}_{n,d}$ *symmetric* if

$$f(x_{\sigma(0)}, \dots, x_{\sigma(n-1)}) = f(x_0, \dots, x_{n-1})$$

for any input $\mathbf{x} = (x_0, \dots, x_{n-1}) \in [d]^n$ and any permutation σ on $[n]$. Define $\mathcal{S}_{n,d}$ as the subset of $\mathcal{F}_{n,d}$ consisting of all symmetric functions.

For a symmetric function $f \in \mathcal{S}_{n,d}$, the value of $f(\mathbf{x})$ is uniquely determined by the *histogram* $\mathbf{h} = (h_0, \dots, h_{d-1})$ of $\mathbf{x}$, where $h_j = |\{i \in [n] : x_i = j\}|$. Formally, we encode $x \in [d]$ into a vector $\phi(x) \in \{0, 1\}^{d-1}$ defined by

$$\phi(x) = \begin{cases} \mathbf{e}_x, & \text{if } x \in \{0, 1, \dots, d-2\}, \\ \mathbf{0}, & \text{if } x = d-1, \end{cases}$$

where $\mathbf{0}$ is the zero vector and $\mathbf{e}_x$ is the unit vector whose entries are 1 at position x and 0 elsewhere. Then, according to [24], there exists a function $f^* : [n+1]^{d-1} \rightarrow \{0, 1\}$ such that for any $(x_0, \dots, x_{n-1}) \in [d]^n$,

$$f(x_0, \dots, x_{n-1}) = f^*(\phi(x_0) + \dots + \phi(x_{n-1})). \quad (8)$$

3.2 Private Simultaneous Messages

A Private Simultaneous Messages (PSM) protocol allows n input parties each holding an input x_i to reveal $f(x_0, \dots, x_{n-1})$ to an external party called a *referee* without revealing any extra information. For convenience, we define PSM protocols in a more general setting where in addition to their private inputs, the parties share a common input z and reveal $f(z; x_0, \dots, x_{n-1})$ to the referee.

Definition 3.1. Let $X_0, \dots, X_{n-1}$, Y , and Z be finite sets. Let $f : Z \times \prod_{i \in [n]} X_i \rightarrow Y$ be a function. An n -party private simultaneous messages (PSM) protocol Π for f is a tuple of algorithms $(\text{Gen}, \text{Enc}, \text{Dec})$, where:

- $\text{Gen}() \rightarrow r$: Gen is a randomized algorithm that takes no input and outputs common randomness $r \in \{0, 1\}^{\text{rand}}$.
- $\text{Enc}(i, (z, x_i), r) \rightarrow m_i$: Enc is a deterministic algorithm that takes an index i , $(z, x_i) \in Z \times X_i$, and $r \in \{0, 1\}^{\text{rand}}$ as input and outputs a message $m_i \in \{0, 1\}^{\text{comm}_i}$.
- $\text{Dec}((m_i)_{i \in [n]}) \rightarrow y$: Dec is a deterministic algorithm that takes $(m_i)_{i \in [n]} \in \prod_{i \in [n]} \{0, 1\}^{\text{comm}_i}$ as input and outputs $y \in Y$.

satisfying the following properties:

Correctness. For any $z \in Z$ and $(x_i)_{i \in [n]} \in \prod_{i \in [n]} X_i$, it holds that

$$\text{Dec}((\text{Enc}(i, (z, x_i), r))_{i \in [n]}) = f(z; x_0, \dots, x_{n-1}),$$

with probability 1 over the choice of $r \leftarrow \text{Gen}()$.

Privacy. There exists a randomized algorithm Sim such that for any $z \in Z$ and $(x_i)_{i \in [n]} \in \prod_{i \in [n]} X_i$,

$$(\text{Enc}(i, (z, x_i), r))_{i \in [n]} \equiv \text{Sim}(f(z; x_0, \dots, x_{n-1})),$$

where the probabilities are taken over $r \leftarrow \text{Gen}()$ and the random coins of Sim .

The communication complexity of Π is defined as $\text{Comm}(\Pi) = \sum_{i \in [n]} \text{comm}_i$. Define $\text{PSM}(f)$ as the minimum of $\text{Comm}(\Pi)$ over all PSM protocols Π for f . For a set $\mathcal{F}$ of functions, define $\text{PSM}(\mathcal{F}) = \max_{f \in \mathcal{F}} \text{PSM}(f)$. The state-of-the-art PSM protocol for general functions is given by [15].

Proposition 3.2 ([15]). $\text{PSM}(\mathcal{F}_{n,d}) = O(d^{n/2}n^3)$.

We suppose that $\text{PSM}(\mathcal{F}_{n,1}) = 0$ (no communication is required as the value of $f : [1]^n \rightarrow \{0, 1\}$ is unique), and $\text{PSM}(\mathcal{F}_{1,d}) = 1$ (since the single party can evaluate a function on his own and send the result to the referee).

We also introduce a variant of PSM protocols for functions that are defined only for a subset of inputs. This variant will be used in an application to ad hoc PSM protocols (see Section 6.1). Let $X_0, \dots, X_{n-1}$, Y , and Z be finite sets. Let $\star$ be a special symbol not belonging to Y . We call a function $f : Z \times \prod_{i \in [n]} X_i \rightarrow Y \cup \{\star\}$ a *partial function* defined over a subset $S \subseteq Z \times \prod_{i \in [n]} X_i$ if $f(z; \mathbf{x}) \in Y$ for all $(z, \mathbf{x}) \in S$ and $f(z; \mathbf{x}) = \star$ for all $(z, \mathbf{x}) \notin S$. Note that $f(z; \mathbf{x}) = \star$ indicates that f is not properly defined over $(z, \mathbf{x})$. We can naturally extend Definition 3.1 to partial functions, where correctness and privacy are required only for inputs in S .

Definition 3.3. Let $X_0, \dots, X_{n-1}$, Y , and Z be finite sets. Let $f : Z \times \prod_{i \in [n]} X_i \rightarrow Y \cup \{\star\}$ be a partial function defined over a subset $S \subseteq Z \times \prod_{i \in [n]} X_i$. An n -party PSM protocol Π for f is a tuple of algorithms $(\text{Gen}, \text{Enc}, \text{Dec})$, where:

- $\text{Gen}() \rightarrow r$: Gen is a randomized algorithm that takes no input and outputs common randomness $r \in \{0, 1\}^{\text{rand}}$.
- $\text{Enc}(i, (z, x_i), r) \rightarrow m_i$: Enc is a deterministic algorithm that takes an index i , $(z, x_i) \in Z \times X_i$, and $r \in \{0, 1\}^{\text{rand}}$ as input and outputs a message $m_i \in \{0, 1\}^{\text{comm}_i}$.
- $\text{Dec}((m_i)_{i \in [n]}) \rightarrow y$: Dec is a deterministic algorithm that takes $(m_i)_{i \in [n]} \in \prod_{i \in [n]} \{0, 1\}^{\text{comm}_i}$ as input and outputs $y \in Y$.

satisfying the following properties:

Correctness. For any $z \in Z$ and $(x_i)_{i \in [n]}$ with $(z, (x_i)_{i \in [n]}) \in S$, it holds that

$$\text{Dec}((\text{Enc}(i, (z, x_i), r))_{i \in [n]}) = f(z; x_0, \dots, x_{n-1}),$$

with probability 1 over the choice of $r \leftarrow \text{Gen}()$.

Privacy. There exists a randomized algorithm Sim such that for any $(z, (x_i)_{i \in [n]}) \in S$,

$$(\text{Enc}(i, x_i, r))_{i \in [n]} \equiv \text{Sim}(f(z; x_0, \dots, x_{n-1})),$$

where the probabilities are taken over $r \leftarrow \text{Gen}()$ and the random coins of Sim .

3.3 Decomposable Affine Randomized Encoding

We introduce a related notion known as Decomposable Affine Randomized Encoding (DARE).

Definition 3.4. Let $\mathbb{F}$ be a finite field and $f : \mathbb{F}^n \rightarrow \mathbb{F}^\ell$ be a function. We say that a function $\hat{f} : \mathbb{F}^n \times \mathbb{F}^m \rightarrow \mathbb{F}^s$ is a Decomposable Affine Randomized Encoding (DARE) of f if it satisfies the following requirements:

Correctness. There exists a function $\text{Dec} : \mathbb{F}^s \rightarrow \mathbb{F}^\ell$, called a decoder, such that for any $x \in \mathbb{F}^n$ and $r \in \mathbb{F}^m$, it holds that $\text{Dec}(\hat{f}(x, r)) = f(x)$.

Privacy. There exists a randomized algorithm Sim such that for any $x \in \mathbb{F}^n$, the distribution of $\text{Sim}(f(x))$ is identical to the distribution of $\hat{f}(x, r)$ induced by $r \leftarrow_{\$} \mathbb{F}^m$.

Decomposable and Affine. It has the form

$$\hat{f}((x_0, \dots, x_{n-1}), r) = (\hat{f}_0(x_0, r), \dots, \hat{f}_{n-1}(x_{n-1}, r))$$

for some functions $\hat{f}_i : \mathbb{F} \times \mathbb{F}^m \rightarrow \mathbb{F}^{s_i}$ such that $\hat{f}_i$ has degree 1 in x_i . That is, $\hat{f}_i(x_i, r)$ can be written as $a_i(r)x_i + b_i(r)$ for arbitrary functions $a_i, b_i : \mathbb{F}^m \rightarrow \mathbb{F}^{s_i}$.

We call s the output length of $\hat{f}$.

It is known that a DARE of a function f can be obtained from an arithmetic formula computing f .

Proposition 3.5 ([30, 20]). If $f : \mathbb{F}^n \rightarrow \mathbb{F}$ is computed by an arithmetic formula of depth d over $\mathbb{F}$, then there exists a DARE of f over $\mathbb{F}$ whose output length is $O_\epsilon(2^{(1+\epsilon)d})$ for any constant $\epsilon > 0$.

3.4 Conditional Disclosure of Secrets

A Conditional Disclosure of Secrets (CDS) protocol assumes n input parties each holding an input x_i and sharing a secret s , and a referee who knows all inputs $(x_i)_{i \in [n]}$. The goal is to reveal s to the referee if and only if $f(x_0, \dots, x_{n-1}) = 1$.

Definition 3.6. Let $X_0, \dots, X_{n-1}$ be finite sets. Let $f : \prod_{i \in [n]} X_i \rightarrow \{0, 1\}$. An n -party conditional disclosure of secrets (CDS) protocol Π for f is a tuple of algorithms $(\text{Gen}, \text{Enc}, \text{Dec})$, where:

- $\text{Gen}() \rightarrow r$: Gen is a randomized algorithm that takes no input and outputs common randomness $r \in \{0, 1\}^{\text{rand}}$.
- $\text{Enc}(i, x_i, s, r) \rightarrow m_i$: Enc is a deterministic algorithm that takes an index i , $x_i \in X_i$, $s \in \{0, 1\}$, and $r \in \{0, 1\}^{\text{rand}}$ as input and outputs a message $m_i \in \{0, 1\}^{\text{comm}_i}$.
- $\text{Dec}((x_i)_{i \in [n]}, (m_i)_{i \in [n]}) \rightarrow s'$: Dec is a deterministic algorithm that takes $(x_i)_{i \in [n]} \in \prod_{i \in [n]} X_i$ and $(m_i)_{i \in [n]} \in \prod_{i \in [n]} \{0, 1\}^{\text{comm}_i}$ as input and outputs $s' \in \{0, 1\}$.

satisfying the following properties:

Correctness. For any $s \in \{0, 1\}$ and any $(x_i)_{i \in [n]} \in \prod_{i \in [n]} X_i$ such that $f((x_i)_{i \in [n]}) = 1$, it holds that

$$\text{Dec}((\text{Enc}(i, f, x_i, s, r))_{i \in [n]}) = s,$$

with probability 1 over the choice of $r \leftarrow \text{Gen}()$.

Privacy. For any $(x_i)_{i \in [n]} \in \prod_{i \in [n]} X_i$ such that $f((x_i)_{i \in [n]}) = 0$, it holds that

$$(\text{Enc}(i, x_i, 0, r))_{i \in [n]} \equiv (\text{Enc}(i, x_i, 1, r'))_{i \in [n]},$$

where the probabilities are taken over independent randomness $r \leftarrow \text{Gen}()$ and $r' \leftarrow \text{Gen}()$.

The communication complexity of Π is defined as $\text{Comm}(\Pi) = \sum_{i \in [n]} \text{comm}_i$. Define $\text{CDS}(f)$ as the minimum of $\text{Comm}(\Pi)$ over all CDS protocols Π for f . For a set $\mathcal{F}$ of functions, define $\text{CDS}(\mathcal{F}) = \max_{f \in \mathcal{F}} \text{CDS}(f)$. The state-of-the-art CDS protocol for general functions is given by [36].

Proposition 3.7 ([36]). $\text{CDS}(\mathcal{F}_{n,d}) = 2^{O(\sqrt{n \log d} \log(n \log d))} = 2^{\tilde{O}(\sqrt{n \log d})}$.

We suppose that $\text{CDS}(\mathcal{F}_{n,1}) = \text{CDS}(\mathcal{F}_{1,d}) = 1$ since in these cases, a single party can decide to send a secret to the referee.

We also need a stronger notion of CDS protocols, referred to as strong CDS protocols [21], in which no information about the secret is leaked as long as at least one of the messages is missing.

Definition 3.8. A CDS protocol $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ for $f : \prod_{i \in [n]} X_i \rightarrow Y$ is said to be strong if for any $\mathbf{x} = (x_i)_{i \in [n]} \in \prod_{i \in [n]} X_i$ and $j \in [n]$, the distribution of a subset $\text{Enc}^{-j}(\mathbf{x}, 0, r)$ of messages is identical to that of $\text{Enc}^{-j}(\mathbf{x}, 1, r)$, where $\text{Enc}^{-j}(\mathbf{x}, s, r) = (\text{Enc}(i, x_i, s, r))_{i \in [n] \setminus \{j\}}$.

It is shown in [21] that any CDS protocol can be upgraded into a strong CDS protocol for the same function by increasing communication complexity only by n bits. Specifically, each party additionally receives a random bit b and additive shares $(b_i)_{i \in [n]}$ of b as common randomness. Then, party i computes a message using a plain CDS protocol with an input x_i and a secret $s + b$. He sends the message along with the i^{th} share of b to the referee. The referee can compute the original secret s only if all messages are obtained and $f((x_i)_{i \in [n]}) = 1$, since only in this case, he can recover both b and $s + b$.

Proposition 3.9 ([21]). For any $f \in \mathcal{F}_{n,d}$, the minimum communication complexity of strong CDS protocols for f is upper bounded by $\text{CDS}(f) + n$.

3.5 Selection Functions

A selection function, denoted by Sel_n , takes as input a bit x_i from each party and a common vector $\mathbf{m}$ of length $n + 1$, and outputs the component of $\mathbf{m}$ corresponding to the sum of the x_i 's. Originally, an efficient PSM protocol for Sel_n was proposed by Benhamouda et al. [18] to upgrade plain PSM protocols into the ones secure against collusion of the referee and parties². Formally, we define a selection function $\text{Sel}_n : \{0, 1\}^{n+1} \times \{0, 1\}^n \rightarrow \{0, 1\}$ as

$$\text{Sel}_n(\mathbf{m}; (x_i)_{i \in [n]}) = \mathbf{m}[s]$$

for any $\mathbf{m} \in \{0, 1\}^{n+1}$ and any $(x_i)_{i \in [n]} \in \{0, 1\}^n$, where $s = \sum_{i \in [n]} x_i$ and $\mathbf{m}[s]$ is the s^{th} element of $\mathbf{m}$. The best known communication complexity for Sel_n is given by [40].

Proposition 3.10 ([40]). $\text{PSM}(\text{Sel}_n) = O(n^2)$.

Note that the protocol in [40] requires computational complexity of $2^{O(n)}$. This drawback is mitigated by the protocol in [18, 23], which has communication complexity $O(n^2 \log n)$ and computational complexity $\tilde{O}(n^2)$. We adopt the protocol in [40] as our primary focus is on communication efficiency.

Next, we consider a weaker variant of Sel_n , denoted by pSel_n , which reveals parties' input bits x_i to the referee while keeping the common string $\mathbf{m}$ hidden. Then, we can obtain a more efficient PSM protocol that achieves the same communication complexity as the protocol in [40], while simultaneously achieving the same computational complexity as [18]. Our construction is inspired by the PSM protocol for a similar functionality in [9]. Roughly speaking, the difference is that the functionality in [9] reveals the component of a common vector $\mathbf{m}$ corresponding to the *product* of the parties' inputs. We define a selection function with public inputs $\text{pSel}_n : \{0, 1\}^{n+1} \times \{0, 1\}^n \rightarrow \{0, 1\}^{n+1}$ as

$$\text{pSel}_n(\mathbf{m}; (x_i)_{i \in [n]}) = (\mathbf{m}[s], x_0, x_1, \dots, x_{n-1})$$

for any $\mathbf{m} \in \{0, 1\}^{n+1}$ and any $(x_i)_{i \in [n]} \in \{0, 1\}^n$, where $s = \sum_{i \in [n]} x_i$ and $\mathbf{m}[s]$ is the s^{th} element of $\mathbf{m}$. In a protocol for pSel_n , we assume that the referee knows the values x_i 's and executes the decoding algorithm using this information. This assumption is satisfied whenever used for constructing CDS protocols. We prove the following proposition in Appendix A.

Proposition 3.11. *There exists a PSM protocol for pSel_n whose communication and computational complexities are $O(n^2)$. In particular, $\text{PSM}(\text{pSel}_n) = O(n^2)$.*

We can straightforwardly extend protocols for Sel_n , $\text{Sel}_{k,n}$, and pSel_n to the setting where each component of $\mathbf{m}$ is an ℓ -bit strings, rather than a single bit, by executing the protocols in a bit-wise manner. The resulting communication complexity increases by only a multiplicative factor of ℓ .

4 Relations to General Functions with Different Domains

In this section, we give a proof of Theorem 1.1. The right inequality of (1) is trivial as $\mathcal{S}_{n,d} \subseteq \mathcal{F}_{n,d}$.

4.1 Upper Bounds: New Constructions of PSM and CDS for $\mathcal{S}_{n,d}$

We prove the upper bound in (2) by showing efficient constructions of PSM and CDS protocols for $\mathcal{S}_{n,d}$. First, we consider the case of PSM.

Proposition 4.1. *Let $f \in \mathcal{S}_{n,d}$ be a symmetric function and $f^* \in \mathcal{F}_{d-1,n+1}$ be a function associated with f satisfying Eq. (8). Then, there exists a PSM protocol Π for f such that $\text{Comm}(\Pi) = \text{PSM}(f^*) \cdot \text{PSM}(\text{Sel}_n)$. In particular, it holds that $\text{PSM}(\mathcal{S}_{n,d}) \leq \text{PSM}(\mathcal{F}_{d-1,n+1}) \cdot \text{PSM}(\text{Sel}_n)$.*

²The original construction of [18] was found to have a flaw by [23] but was fixed without compromising efficiency.

Notations.

- Let $\Pi_{f^*} = (\text{Gen}_{f^*}, \text{Enc}_{f^*}, \text{Dec}_{f^*})$ be a $(d-1)$ -party PSM protocol for f^* .
- Let $\Pi_{\text{sel}} = (\text{Gen}_{\text{sel}}, \text{Enc}_{\text{sel}}, \text{Dec}_{\text{sel}})$ be an n -party PSM protocol for Sel_n .

$\text{Gen}()$.

1. Run $r_{f^*} \leftarrow \text{Gen}_{f^*}()$.
2. For each $k \in [d-1]$, run $r_{\text{sel}}^{(k)} \leftarrow \text{Gen}_{\text{sel}}()$ with independent randomness.
3. For each $k \in [d-1]$ and each $j \in [n+1]$, compute a message $s_j^{(k)} = \text{Enc}_{f^*}(k, j, r_{f^*})$ of the k^{th} party in Π_g encoding an input $j \in [n+1]$.
4. Output $R = ((s^{(k)})_{k \in [d-1]}, (r_{\text{sel}}^{(k)})_{k \in [d-1]})$, where $s^{(k)} = (s_j^{(k)})_{j \in [n+1]}$.

$\text{Enc}(i, x_i, R)$.

1. Let $\phi(x_i) = (v_i^{(0)}, \dots, v_i^{(d-2)}) \in \{0, 1\}^{d-1}$.
2. For each $k \in [d-1]$, compute $m_i^{(k)} = \text{Enc}_{\text{sel}}(i, (s^{(k)}, v_i^{(k)}), r_{\text{sel}}^{(k)})$.
3. Output $M_i = (m_i^{(k)})_{k \in [d-1]}$.

$\text{Dec}(M_0, \dots, M_{n-1})$.

1. For each $k \in [d-1]$, compute $t^{(k)} = \text{Dec}_{\text{sel}}(m_0^{(k)}, \dots, m_{n-1}^{(k)})$.
2. Output $y = \text{Dec}_{f^*}(t^{(0)}, \dots, t^{(d-2)})$.

Figure 1: A PSM protocol for $\mathcal{S}_{n,d}$

Proof. Let Π_{f^*} be a $(d-1)$ -party PSM protocol for f^* and Π_{sel} be an n -party PSM protocol for Sel_n . We describe a PSM protocol Π for f in Figure 1.

Correctness. Let $(x_0, \dots, x_{n-1}) \in [d]^n$ be inputs. Let $(h^{(k)})_{k \in [d-1]} = \sum_{i \in [n]} \phi(x_i) \in [n+1]^{d-1}$. For each $k \in [d-1]$, it follows from the correctness of Π_{sel} that

$$t^{(k)} = \text{Sel}_n((s^{(k)}, v_0^{(k)}), \dots, (s^{(k)}, v_{n-1}^{(k)})) = \text{Enc}_{f^*}(k, h^{(k)}, r_{f^*}).$$

Thus, it holds that $y = f^*(h^{(0)}, \dots, h^{(d-2)}) = f(x_0, \dots, x_{n-1})$.

Privacy. Let $(x_0, \dots, x_{n-1}) \in [d]^n$ be inputs and $(h^{(k)})_{k \in [d-1]} = \sum_{i \in [n]} \phi(x_i) \in [n+1]^{d-1}$. Let Sim_{f^*} be a simulator for Π_{f^*} and Sim_{sel} be a simulator for Π_{sel} . For each $k \in [d-1]$, the messages $m_i^{(k)}$'s can be simulated by Sim_{sel} with input $\text{Sel}_n((s^{(k)}, v_0^{(k)}), \dots, (s^{(k)}, v_{n-1}^{(k)}))$, that is, $(m_i^{(k)})_{i \in [n]} \equiv \text{Sim}_{\text{sel}}(s_{h^{(k)}}^{(k)})$. Since $s_{h^{(k)}}^{(k)} = \text{Enc}_{f^*}(k, h^{(k)}, r_{f^*})$, the $s_{h^{(k)}}^{(k)}$'s can be simulated by Sim_{f^*} with input $g(h^{(0)}, \dots, h^{(d-2)})$, that is, $(s_{h^{(k)}}^{(k)})_{k \in [d-1]} \equiv \text{Sim}_{f^*}(f^*(h^{(0)}, \dots, h^{(d-2)}))$. Therefore, since $f^*(h^{(0)}, \dots, h^{(d-2)}) = f(x_0, \dots, x_{n-1})$, we obtain a simulator simulating $(M_i)_{i \in [n]} = (m_i^{(k)})_{k \in [d-1], i \in [n]}$ from $f(x_0, \dots, x_{n-1})$.

Complexity. We have that $|s_j^{(k)}| \leq \text{comm}_k(\Pi_{f^*})$, where $\text{comm}_k(\Pi_{f^*})$ denote the bit length of messages of the k^{th} party in Π_{f^*} . We have that $\sum_{i \in [n]} |m_i^{(k)}| \leq \text{PSM}(\text{Sel}_n) \cdot \text{comm}_k(\Pi_{f^*})$ for each $k \in [d-1]$. Thus, we obtain that $\sum_{i \in [n]} |M_i| \leq \text{PSM}(\text{Sel}_n) \sum_{k \in [d-1]} \text{comm}_k(\Pi_{f^*}) \leq \text{PSM}(f^*) \cdot \text{PSM}(\text{Sel}_n)$. $\square$

By plugging in Propositions 3.2 and 3.10, we obtain the following corollary. In particular, this improves the prior upper bound of $n^{2d/3+O(1)}$ in [24].

Corollary 4.2. $\text{PSM}(\mathcal{S}_{n,d}) \leq \text{PSM}(\mathcal{F}_{d-1,n+1}) \cdot O(n^2) \leq O((n+1)^{d/2} n^{3/2} d^3)$.

Notations.

- Let $\Pi_{f^*} = (\text{Gen}_{f^*}, \text{Enc}_{f^*}, \text{Dec}_{f^*})$ be a $(d-1)$ -party CDS protocol for f^* .
- Let $\Pi_{\text{psel}} = (\text{Gen}_{\text{psel}}, \text{Enc}_{\text{psel}}, \text{Dec}_{\text{psel}})$ be an n -party PSM protocol for pSel_n .

$\text{Gen}()$.

1. Run $r_{f^*} \leftarrow \text{Gen}_{f^*}()$.
2. For each $k \in [d-1]$, run $r_{\text{sel}}^{(k)} \leftarrow \text{Gen}_{\text{psel}}()$ with independent randomness.
3. For each $k \in [d-1]$ and each $j \in [n+1]$, compute a message $l_j^{(k)} = \text{Enc}_{f^*}(k, j, 0, r_{f^*})$ of the k^{th} party in Π_{f^*} encoding an input $j \in [n+1]$ and a secret $s = 0$.
4. For each $k \in [d-1]$ and each $j \in [n+1]$, compute a message $m_j^{(k)} = \text{Enc}_{f^*}(k, j, 1, r_{f^*})$ of the k^{th} party in Π_{f^*} encoding an input $j \in [n+1]$ and a secret $s = 1$.
5. Output $R = ((\mathbf{l}^{(k)})_{k \in [d-1]}, (\mathbf{m}^{(k)})_{k \in [d-1]}, (r_{\text{sel}}^{(k)})_{k \in [d-1]})$, where $\mathbf{l}^{(k)} = (l_j^{(k)})_{j \in [n+1]}$ and $\mathbf{m}^{(k)} = (m_j^{(k)})_{j \in [n+1]}$.

$\text{Enc}(i, x_i, s, R)$.

1. Let $\phi(x_i) = (v_i^{(0)}, \dots, v_i^{(d-2)}) \in \{0, 1\}^{d-1}$.
2. If $s = 0$: For each $k \in [d-1]$, compute $\mu_i^{(k)} = \text{Enc}_{\text{psel}}(i, (\mathbf{l}^{(k)}, v_i^{(k)}), r_{\text{psel}}^{(k)})$.
3. If $s = 1$: For each $k \in [d-1]$, compute $\mu_i^{(k)} = \text{Enc}_{\text{psel}}(i, (\mathbf{m}^{(k)}, v_i^{(k)}), r_{\text{psel}}^{(k)})$.
4. Output $M_i = (\mu_i^{(k)})_{k \in [d-1]}$.

$\text{Dec}((x_i)_{i \in [n]}, (M_i)_{i \in [n]})$.

1. Let $\mathbf{v}_i = (v_i^{(k)})_{k \in [d-1]} = \phi(x_i)$ for each $i \in [n]$ and compute $(h^{(k)})_{k \in [d-1]} = \sum_{i \in [n]} \mathbf{v}_i \in [n+1]^{d-1}$.
2. For each $k \in [d-1]$, compute $t^{(k)} = \text{Dec}_{\text{psel}}(\mu_0^{(k)}, \dots, \mu_{n-1}^{(k)})$ with the knowledge of $(v_i^{(k)})_{i \in [n]}$.
3. Output $y = \text{Dec}_{f^*}((h^{(k)})_{k \in [d-1]}, (t^{(k)})_{k \in [d-1]})$.

Figure 2: A CDS protocol for $\mathcal{S}_{n,d}$

Next, in the case of CDS, the following proposition holds.

Proposition 4.3. *Let $f \in \mathcal{S}_{n,d}$ be a symmetric function and $f^* \in \mathcal{F}_{d-1,n+1}$ be a function associated with f satisfying Eq. (8). Then, there exists a CDS protocol Π for f such that $\text{Comm}(\Pi) = \text{CDS}(f^*) \cdot \text{PSM}(\text{pSel}_n)$. In particular, it holds that $\text{CDS}(\mathcal{S}_{n,d}) \leq \text{CDS}(\mathcal{F}_{d-1,n+1}) \cdot \text{PSM}(\text{pSel}_n)$.*

Proof. Let Π_{f^*} be a $(d-1)$ -party CDS protocol for f^* and Π_{psel} be an n -party PSM protocol for pSel_n . We describe a CDS protocol Π for f in Figure 2.

Correctness. Let $(x_0, \dots, x_{n-1}) \in [d]^n$ be inputs such that $f(x_0, \dots, x_{n-1}) = 1$. Let $(h^{(k)})_{k \in [d-1]} = \sum_{i \in [n]} \phi(x_i) \in [n+1]^{d-1}$. For each $k \in [d-1]$, it holds that

$$t^{(k)} = \begin{cases} l_{h^{(k)}}^{(k)}, & \text{if } s = 0, \\ m_{h^{(k)}}^{(k)}, & \text{if } s = 1 \end{cases} = \text{Enc}_{f^*}(k, h^{(k)}, s, r_{f^*}).$$

Since $f^*(h^{(0)}, \dots, h^{(d-2)}) = f(x_0, \dots, x_{n-1}) = 1$, the correctness of Π_{f^*} implies that

$$y = \text{Dec}_{f^*}((h^{(k)})_{k \in [d-1]}, (t^{(k)})_{k \in [d-1]}) = s.$$

Privacy. Let $(x_0, \dots, x_{n-1}) \in [d]^n$ be inputs with $f(x_0, \dots, x_{n-1}) = 0$. Let $(h^{(k)})_{k \in [d-1]} = \sum_{i \in [n]} \phi(x_i) \in [n+1]^{d-1}$. Let Sim_{psel} be a simulator for Π_{psel} . For each $k \in [d-1]$, if $s = 0$, the messages $\mu_i^{(k)}$'s can be simulated by Sim_{psel} with input $\text{pSel}_n((l^{(k)}, v_0^{(k)}), \dots, (l^{(k)}, v_{n-1}^{(k)}))$, that is, $(\mu_i^{(k)})_{i \in [n]} \equiv \text{Sim}_{\text{psel}}((v_i^{(k)})_{i \in [n]}, l_{h^{(k)}}^{(k)})$. Similarly, if $s = 1$, it holds that $(\mu_i^{(k)})_{i \in [n]} \equiv \text{Sim}_{\text{psel}}((v_i^{(k)})_{i \in [n]}, m_{h^{(k)}}^{(k)})$. On the other hand, since $f^*(h^{(0)}, \dots, h^{(d-2)}) = f(x_0, \dots, x_{n-1}) = 0$, the privacy of Π_{f^*} implies that $(l_{h^{(k)}}^{(k)})_{k \in [d-1]} \equiv (m_{h^{(k)}}^{(k)})_{k \in [d-1]}$. Therefore, since the $v_i^{(k)}$'s depend solely on the x_i 's, we conclude that the distribution of the messages $(M_i)_{i \in [n]}$ is independent of a secret s .

Complexity. We have that $|l_j^{(k)}| \leq \text{comm}_k(\Pi_{f^*})$ and $|m_j^{(k)}| \leq \text{comm}_k(\Pi_{f^*})$, where $\text{comm}_k(\Pi_{f^*})$ denote the bit length of messages of the k^{th} party in Π_{f^*} . We thus have that $\sum_{i \in [n]} |\mu_i^{(k)}| \leq \text{PSM}(\text{pSel}_n) \cdot \text{comm}_k(\Pi_{f^*})$ for each $k \in [d-1]$. Thus, we obtain that $\sum_{i \in [n]} |M_i| \leq \text{PSM}(\text{pSel}_n) \cdot \sum_{k \in [d-1]} \text{comm}_k(\Pi_{f^*}) \leq \text{CDS}(f^*) \cdot \text{PSM}(\text{pSel}_n)$. $\square$

By plugging in Propositions 3.7 and 3.11, we obtain the following corollary.

Corollary 4.4. $\text{CDS}(\mathcal{S}_{n,d}) \leq \text{CDS}(\mathcal{F}_{d-1,n+1}) \cdot O(n^2) \leq n^2 2^{\tilde{O}(\sqrt{d \log n})}$.

Prior to this work, the only known bound for $\text{CDS}(\mathcal{S}_{n,d})$ was the trivial one $\text{CDS}(\mathcal{F}_{n,d})$. The currently best known CDS protocol for general functions requires sub-exponential communication in n [36], whereas our protocol achieves polynomial communication when $d = o(\log n)$. This is the first result to demonstrate that the symmetry of functions is effective to reduce the complexity of CDS.

4.2 Lower Bounds

We prove the lower bounds in (1) and (2).

Proposition 4.5. For $\mathfrak{C} \in \{\text{PSM}, \text{CDS}\}$, $\mathfrak{C}(\mathcal{S}_{n,d}) \geq \mathfrak{C}(\mathcal{F}_{n,d'})$, where $d' = \lfloor d/n \rfloor$.

Proof. Let $g : [d']^n \rightarrow \{0, 1\}$ be a function in $\mathcal{F}_{n,d'}$. Let $S = [n] \times [d']$ and T be a set of size $d - nd'$ that is disjoint from S . Let $X = S \cup T$. We define a symmetric function $f : X^n \rightarrow \{0, 1\}$ as follows. Let $(a_0, \dots, a_{n-1}) \in X^n$. If $a_k = (i_k, j_k) \in S$ for all $k \in [n]$ and the i_k 's are pairwise distinct, then sort $(a_0, \dots, a_{n-1})$ in lexicographical order, i.e., according to the values of $(i_0, \dots, i_{n-1})$. Let $((0, \ell_0), \dots, (n-1, \ell_{n-1}))$ denote the sorted array. We define $f(a_0, \dots, a_{n-1}) := g(\ell_0, \dots, \ell_{n-1})$. Otherwise, we set the value of f to 0. Clearly, $f \in \mathcal{S}_{n,d}$ as it sorts inputs before computing an output.

Given any PSM protocol $\Pi_f = (\text{Gen}_f, \text{Enc}_f, \text{Dec}_f)$ for f , we construct a PSM protocol $\Pi_g = (\text{Gen}_g, \text{Enc}_g, \text{Dec}_g)$ for g with the same communication complexity, which proves $\text{PSM}(\mathcal{S}_{n,d}) \geq \text{PSM}(\mathcal{F}_{n,d'})$. For $i \in [n]$, let P_i and Q_i be the i^{th} party involved in Π_g and Π_f , respectively. Π_g proceeds as follows:

Gen_g(): Run $r_f \leftarrow \text{Gen}_f()$ and output $r_g := r_f$.

Enc_g(i, x_i, r_g): P_i emulates the roles of Q_i in Π_f with input $a_i = (i, x_i)$. That is, P_i computes $M_i = \text{Enc}_f(i, a_i, r_f)$.

Dec_g($M_0, \dots, M_{n-1}$): Output $y = \text{Dec}_f(M_0, \dots, M_{n-1})$.

The correctness follows from the fact that $y = f((0, x_0), \dots, (n-1, x_{n-1})) = g(x_0, \dots, x_{n-1})$. The privacy directly follows from that of Π_f since what the referee learns is exactly messages resulting from an execution of Π_f .

The statements for CDS follow from the same argument except that the PSM protocols Π_f and Π_g are replaced with CDS protocols for f and g , respectively. $\square$

Proposition 4.6. For $\mathfrak{C} \in \{\text{PSM}, \text{CDS}\}$, $\mathfrak{C}(\mathcal{S}_{n,d}) \geq \mathfrak{C}(\mathcal{F}_{d-1,n'+1})$, where $n' = \lfloor n/(d-1) \rfloor$.

Proof. Let $g : [n'+1]^{d-1} \rightarrow \{0, 1\}$ be a function in $\mathcal{F}_{d-1,n'+1}$. We define a symmetric function $f : [d]^n \rightarrow \{0, 1\}$ as follows. Due to the symmetry, it suffices to specify all values of f for inputs sorted in ascending order. For any $(x_0, \dots, x_{d-2}) \in [n'+1]^{d-1}$, define

$$f(\underbrace{0, \dots, 0}_{x_0}, \dots, \underbrace{d-2, \dots, d-2}_{x_{d-2}}, \underbrace{d-1, \dots, d-1}_{n - \sum_{i \in [d-1]} x_i}) = g(x_0, \dots, x_{d-2}).$$

Note that $n - \sum_{i \in [d-1]} x_i \geq 0$ since $n \geq n'(d-1)$. For other inputs, set the value of f to an arbitrary value (say, 0). Let $\delta = n - n'(d-1)$.

We show that given any PSM protocol $\Pi_f = (\text{Gen}_f, \text{Enc}_f, \text{Dec}_f)$ for f , it is possible to obtain a PSM protocol $\Pi_g = (\text{Gen}_g, \text{Enc}_g, \text{Dec}_g)$ for g with the same communication complexity, which proves $\text{PSM}(\mathcal{S}_{n,d}) \geq \text{PSM}(\mathcal{F}_{d-1,n'+1})$. For $j \in [d-1]$, let P_j be the j^{th} party involved in Π_g . We partition the set of n parties involved in Π_f , into $d-1$ subsets $S_0, \dots, S_{d-2}$ such that $|S_j| = n'$ for all $0 \leq j \leq d-3$ and $|S_{d-2}| = n - n'(d-2) = n' + \delta$. We denote $S_j = \{Q_{j,i} : i \in [n']\}$ for $0 \leq j \leq d-3$ and $S_{d-2} = \{Q_{d-2,i} : i \in [n' + \delta]\}$. Π_g proceeds as follows:

Gen_g($\cdot$): Run $r_f \leftarrow \text{Gen}_f(\cdot)$ and output $r_g := r_f$.

Enc_g(j, x_j, r_g):

- If $0 \leq j \leq d-3$, then P_j emulates the roles of parties in S_j by computing

$$M_j = (\mu_{j,0}(j), \dots, \mu_{j,x_j-1}(j), \mu_{j,x_j}(d-1), \dots, \mu_{j,n'-1}(d-1)),$$

where $\mu_{j,i}(x)$ denotes a message of $Q_{j,i}$ with input x and randomness r_f .

- If $j = d-2$, then P_{d-2} emulates the roles of parties in S_{d-2} by computing

$$M_{d-2} = (\mu_{d-2,0}(d-2), \dots, \mu_{d-2,x_{d-2}-1}(d-2), \mu_{d-2,x_{d-2}}(d-1), \dots, \mu_{d-2,n'-1+\delta}(d-1)).$$

Dec_g($M_0, \dots, M_{d-2}$):

1. Parse M_j into $M_j = (M_{j,i})_{i \in [n']}$ for $j \in [d-2]$ and $M_{d-2} = (M_{d-2,i})_{i \in [n'+\delta]}$.
2. Output $y = \text{Dec}_f((M_{j,i})_{j \in [d-2], i \in [n]}, (M_{d-2,i})_{i \in [n'+\delta]})$.

The correctness follows since $\sum_{j \in [d-2]} (n' - x_j) + (n' + \delta - x_{d-2}) = n - \sum_{j \in [d-1]} x_j$ and

$$y = f(\underbrace{0, \dots, 0}_{x_0}, \dots, \underbrace{d-2, \dots, d-2}_{x_{d-2}}, \underbrace{d-1, \dots, d-1}_{n - \sum_{j \in [d-1]} x_j}) = g(x_0, \dots, x_{d-2}).$$

The second equality follows from the symmetry of f . The privacy directly follows from that of Π_f since what the referee learns is exactly messages resulting from an execution of Π_f .

The statements for CDS follow from the same argument except that the PSM protocols Π_f and Π_g are replaced with CDS protocols for f and g , respectively. $\square$

5 Towards Tight Relations to General Functions

In this section, we give a proof of Theorem 1.2. To this end, we relate the optimal communication complexities of general functions with different domain sizes. Specifically, it would be desirable to obtain an inequality of the form $\mathfrak{C}(\mathcal{F}_{n,d_2}) \leq M \cdot \mathfrak{C}(\mathcal{F}_{n,d_1})$ for $d_1 < d_2$ and $\mathfrak{C} \in \{\text{PSM}, \text{CDS}\}$. Technically, we present techniques for constructing PSM and CDS protocols for $\mathcal{F}_{n,d_2}$ based on those for $\mathcal{F}_{n,d_1}$.

5.1 The Case of CDS

We begin with the case of CDS, since it is technically easier to handle.

We prepare some notations. Define $\delta_w(y) = 1$ if $y = w$ and 0 if $y \neq w$. For each $\mathbf{w} = (w_i)_{i \in [n]} \in \{0,1\}^n$, define $\delta_{\mathbf{w}} : \{0,1\}^n \rightarrow \{0,1\}$ as $\delta_{\mathbf{w}}((y_i)_{i \in [n]}) = \prod_{i \in [n]} \delta_{w_i}(y_i)$. Let $F \in \mathcal{F}_{n,2d}$. We interpret each input $x' \in [2d]$ as a pair (x, y) where $x \in [d]$ and $y \in \{0,1\}$. For each $\mathbf{w} \in \{0,1\}^n$, let $f_{\mathbf{w}} \in \mathcal{F}_{n,d}$ denote the function obtained by fixing each y -component to w_i , that is, $f_{\mathbf{w}}((x_i)_{i \in [n]}) = F((x_i, w_i)_{i \in [n]})$. Furthermore, for a function $f \in \mathcal{F}_{n,d}$, we define a function $G_f \in \mathcal{F}_{n,2d}$ with extended domains as

$$G_f((x_i, y_i)_{i \in [n]}) = f((x_i)_{i \in [n]}) \prod_{i \in [n]} y_i$$

for any $x_i \in [d]$ and $y_i \in \{0,1\}$.

First, we reduce a CDS protocol for $F \in \mathcal{F}_{n,2d}$ to CDS protocols for G_f with $f \in \mathcal{F}_{n,d}$.

Notations.

- For $f \in \mathcal{F}_{n,d}$, let $\Pi_{G_f} = (\text{Gen}_{G_f}, \text{Enc}_{G_f}, \text{Dec}_{G_f})$ be a CDS protocol for G_f .

$\text{Gen}()$.

1. For each $\mathbf{w} \in \{0,1\}^n$, run $r_{\mathbf{w}} \leftarrow \text{Gen}_{G_{f_{\mathbf{w}}}}()$.
2. Output $R = (r_{\mathbf{w}})_{\mathbf{w} \in \{0,1\}^n}$.

$\text{Enc}(i, (x_i, y_i), s, R)$.

1. For each $\mathbf{w} = (w_i)_{i \in [n]} \in \{0,1\}^n$, compute $m_{\mathbf{w},i} = \text{Enc}_{G_{f_{\mathbf{w}}}}(i, (x_i, \delta_{w_i}(y_i)), s, r_{\mathbf{w}})$.
2. Output $M_i = (m_{\mathbf{w},i})_{\mathbf{w} \in \{0,1\}^n}$.

$\text{Dec}((x_i, y_i)_{i \in [n]}; M_0, \dots, M_{n-1})$.

1. If $F((x_i, y_i)_{i \in [n]}) = 0$, then output $z = \perp$.
2. If $F((x_i, y_i)_{i \in [n]}) = 1$, then output $z = \text{Dec}_{G_{f_{\mathbf{w}}}}((x_i, 1)_{i \in [n]}; (m_{\mathbf{y},i})_{i \in [n]})$.

Figure 3: A CDS protocol for F

Lemma 5.1. $\text{CDS}(\mathcal{F}_{n,2d}) \leq 2^n \cdot \max_{f \in \mathcal{F}_{n,d}} \text{CDS}(G_f)$.

Proof. Let $F \in \mathcal{F}_{n,2d}$. Suppose that for any $f \in \mathcal{F}_{n,d}$, we are given a CDS protocol $\Pi_{G_f} = (\text{Gen}_{G_f}, \text{Enc}_{G_f}, \text{Dec}_{G_f})$ for G_f . We show a CDS protocol Π_F for F using the Π_{G_f} 's in Figure 3.

First, we see the correctness of Π_F . Let $\mathbf{x} = (x_i)_{i \in [n]} \in [d]^n$ and $\mathbf{y} = (y_i)_{i \in [n]} \in \{0,1\}^n$ be inputs such that $F((x_i, y_i)_{i \in [n]}) = 1$ and $s \in \{0,1\}$ be a secret. Since

$$F((x_i, y_i)_{i \in [n]}) = \sum_{\mathbf{w} \in \{0,1\}^n} f_{\mathbf{w}}(\mathbf{x}) \delta_{\mathbf{w}}(\mathbf{y}),$$

we have that $G_{f_{\mathbf{y}}}((x_i, \delta_{y_i}(y_i))_{i \in [n]}) = f_{\mathbf{y}}(\mathbf{x}) \delta_{\mathbf{y}}(\mathbf{y}) = 1$ and $G_{f_{\mathbf{w}}}((x_i, \delta_{w_i}(y_i))_{i \in [n]}) = 0$ for $\mathbf{w} \neq \mathbf{y}$. Thus, it follows from the correctness of $\Pi_{G_{f_{\mathbf{y}}}}$ that $z = \text{Dec}_{G_{f_{\mathbf{y}}}}((x_i, \delta_{y_i}(y_i))_{i \in [n]}; (m_{\mathbf{y},i})_{i \in [n]}) = s$.

Next, we see the privacy of Π_F . Let $\mathbf{x} = (x_i)_{i \in [n]} \in [d]^n$ and $\mathbf{y} = (y_i)_{i \in [n]} \in \{0,1\}^n$ be inputs such that $F((x_i, y_i)_{i \in [n]}) = 0$. For any $\mathbf{w} \in \{0,1\}^n$, we have that

$$G_{f_{\mathbf{w}}}((x_i, \delta_{w_i}(y_i))_{i \in [n]}) = \begin{cases} f_{\mathbf{y}}(\mathbf{x}) \delta_{\mathbf{y}}(\mathbf{y}) = 0, & \text{if } \mathbf{w} = \mathbf{y}, \\ f_{\mathbf{w}}(\mathbf{x}) \delta_{\mathbf{w}}(\mathbf{y}) = 0, & \text{if } \mathbf{w} \neq \mathbf{y}. \end{cases}$$

Thus, the privacy of each $\Pi_{G_{f_{\mathbf{w}}}}$ ensures that the distribution of $(m_{\mathbf{w},i})_{i \in [n]}$ when $s = 0$ is identical to that when $s = 1$. Since all $\Pi_{G_{f_{\mathbf{w}}}}$ are executed on independent randomness, the distribution of messages $(m_{\mathbf{w},i})_{i \in [n], \mathbf{w} \in \{0,1\}^n}$ when $s = 0$ is identical to that when $s = 1$.

Since $f_{\mathbf{w}} \in \mathcal{F}_{n,d}$, the communication complexity of each $\Pi_{G_{f_{\mathbf{w}}}}$ is upper bounded by $\max_{f \in \mathcal{F}_{n,d}} \text{CDS}(G_f)$. Since Π_F internally executes 2^n such protocols, the complexity of Π_F is upper bounded by $2^n \cdot \max_{f \in \mathcal{F}_{n,d}} \text{CDS}(G_f)$. $\square$

Next, we reduce a CDS protocol for G_f with $f \in \mathcal{F}_{n,d}$ to a strong CDS for f . Proposition 3.9 ensures that the strong CDS is efficiently reduced to a plain CDS for f .

Lemma 5.2. For any $f \in \mathcal{F}_{n,d}$, $\text{CDS}(G_f) \leq \text{CDS}(f) + n$.

Proof. Let $f \in \mathcal{F}_{n,d}$ and suppose that a strong CDS protocol Π_f for f is given. We show a CDS protocol Π_{G_f} for G_f in Figure 4.

Notations.

- Let $\Pi_f = (\text{Gen}_f, \text{Enc}_f, \text{Dec}_f)$ be a strong CDS protocol for f .

$\text{Gen}()$. Output $r \leftarrow \text{Gen}_f()$.

$\text{Enc}(i, (x_i, y_i), s, r)$.

1. If $y_i = 0$, output $m_i = \perp$.
2. If $y_i = 1$, output $m_i = \text{Enc}_f(i, x_i, s, r)$.

$\text{Dec}((x_i, y_i)_{i \in [n]}; M_0, \dots, M_{n-1})$.

1. If $y_i = 1$ for all i and $f((x_i)_{i \in [n]}) = 1$, output $z = \text{Dec}_f((x_i)_{i \in [n]}; (m_i)_{i \in [n]})$.
2. Otherwise, output $z = \perp$.

Figure 4: A CDS protocol for G_f

First, we see the correctness of Π_{G_f} . Let $\mathbf{x} = (x_i)_{i \in [n]} \in [d]^n$ and $\mathbf{y} = (y_i)_{i \in [n]} \in \{0, 1\}^n$ be inputs such that $G_f((x_i, y_i)_{i \in [n]}) = 1$, and $s \in \{0, 1\}$ be a secret. In particular, $y_i = 1$ for all $i \in [n]$ and $f(\mathbf{x}) = 1$. Thus, it follows from the correctness of Π_f that $z = \text{Dec}_f(\mathbf{x}; (m_i)_{i \in [n]}) = s$.

Next, we see the privacy of Π_{G_f} . Let $\mathbf{x} = (x_i)_{i \in [n]} \in [d]^n$ and $\mathbf{y} = (y_i)_{i \in [n]} \in \{0, 1\}^n$ be inputs such that $G_f((x_i, y_i)_{i \in [n]}) = 0$. First, suppose that $y_i = 1$ for all $i \in [n]$. This implies $f(\mathbf{x}) = 0$. Thus, the privacy of Π_f ensures that the distribution of messages $(\text{Enc}_f(i, x_i, 0, r))_{i \in [n]}$ is identical to that of $(\text{Enc}_f(i, x_i, 1, r))_{i \in [n]}$. Second, suppose that $y_i = 0$ for some $i \in [n]$. Since Π_f is a strong CDS protocol, the distribution of $(\text{Enc}_f(i', x_{i'}, 0, r))_{i' \in [n] \setminus \{i\}}$ is identical to that of $(\text{Enc}_f(i', x_{i'}, 1, r))_{i' \in [n] \setminus \{i\}}$. Thus, the distribution of messages are independent of a secret.

Clearly, the communication complexity of Π_{G_f} is at most that of Π_f . Thus, it follows from Proposition 3.9 that $\text{CDS}(\Pi_{G_f}) \leq \text{Comm}(\Pi_f) \leq \text{CDS}(f) + n$. $\square$

Combining Lemmas 5.1 and 5.2, we obtain the following inequalities.

Lemma 5.3. $\text{CDS}(\mathcal{F}_{n,2d}) \leq 2^n (\text{CDS}(\mathcal{F}_{n,d}) + n) \leq n2^n \cdot \text{CDS}(\mathcal{F}_{n,d})$.

By applying Lemma 5.3 iteratively, we obtain the following proposition.

Proposition 5.4. For $d_1 \leq d_2$, $\text{CDS}(\mathcal{F}_{n,d_2}) \leq (O(d_2/d_1))^{n+\log n} \cdot \text{CDS}(\mathcal{F}_{n,d_1})$.

Proof. Set $\ell = \lceil \log(d_2/d_1) \rceil$. It holds that

$$\text{CDS}(\mathcal{F}_{n,d_2}) \leq 2^n n \cdot \text{CDS}(\mathcal{F}_{n,d_2/2}) \leq \dots \leq (2^n n)^\ell \cdot \text{CDS}(\mathcal{F}_{n,2^{-\ell}d_2}).$$

Since $2^{-\ell}d_2 \leq d_1$ and $(2^n n)^\ell = (O(d_2/d_1))^{n+\log n}$, the stated inequality is obtained. $\square$

Now, we give a proof of Theorem 1.2 in the case of $\mathfrak{C} = \text{CDS}$. From Theorem 1.1, we have an inequality $\text{CDS}(\mathcal{S}_{n,d}) \geq \text{CDS}(\mathcal{F}_{n,d'})$ for $d' = \lfloor d/n \rfloor$. Since $d/d' = O(n)$, it follows from Proposition 5.4 that $\text{CDS}(\mathcal{F}_{n,d}) \leq n^{O(n)} \cdot \text{CDS}(\mathcal{F}_{n,d'})$, which proves the lower bound in (3). On the other hand, we have an inequality $\text{CDS}(\mathcal{S}_{n,d}) \geq \text{CDS}(\mathcal{F}_{d-1,n'+1})$ for $n' = \lfloor n/(d-1) \rfloor$. Since $n/n' = O(d)$, it again follows from Proposition 5.4 that $\text{CDS}(\mathcal{F}_{d-1,n'+1}) \leq d^{O(d)} \cdot \text{CDS}(\mathcal{F}_{d-1,n'+1})$, which proves the lower bound in (4).

5.2 The Case of PSM

Next, we give a proof of Theorem 1.2 in the case of $\mathfrak{C} = \text{PSM}$. We prepare some additional notations. For a class $\mathcal{F}$ of functions with domain $\prod_{i \in [n]} X_i$, we introduce a class $\mathcal{F}^{\text{plus}}$ that augments $\mathcal{F}$ with an additional bit. Specifically, for $f : \prod_{i \in [n]} X_i \rightarrow \{0, 1\} \in \mathcal{F}$, we define a function $f^{\text{plus}} : \{0, 1\} \times \prod_{i \in [n]} X_i \rightarrow \{0, 1\}$ as

$$f^{\text{plus}}(b; (x_i)_{i \in [n]}) = f((x_i)_{i \in [n]}) + b \bmod 2$$

Notations.

- For $f \in \mathcal{F}_{n,d}$, let $\Pi_{G_f^{\text{plus}}} = (\text{Gen}_{G_f^{\text{plus}}}, \text{Enc}_{G_f^{\text{plus}}}, \text{Dec}_{G_f^{\text{plus}}})$ be a PSM protocol for G_f^{plus} .

$\text{Gen}()$.

1. For each $\mathbf{w} \in \{0,1\}^n$, run $r_{\mathbf{w}} \leftarrow \text{Gen}_{G_{f_{\mathbf{w}}}^{\text{plus}}}()$.
2. Choose $b_{\mathbf{w}} \in \{0,1\}$ for each $\mathbf{w} \in \{0,1\}^n$ uniformly at random conditioned on $\sum_{\mathbf{w} \in \{0,1\}^n} b_{\mathbf{w}} = 0$.
3. Output $R = ((r_{\mathbf{w}})_{\mathbf{w} \in \{0,1\}^n}, (b_{\mathbf{w}})_{\mathbf{w} \in \{0,1\}^n})$.

$\text{Enc}(i, (x_i, y_i), R)$.

1. For each $\mathbf{w} = (w_i)_{i \in [n]} \in \{0,1\}^n$, compute $m_{\mathbf{w},i} = \text{Enc}_{G_{f_{\mathbf{w}}}^{\text{plus}}}(i, (b_{\mathbf{w}}, x_i, \delta_{w_i}(y_i)), r_{\mathbf{w}})$.
2. Output $M_i = (m_{\mathbf{w},i})_{\mathbf{w} \in \{0,1\}^n}$.

$\text{Dec}(M_0, \dots, M_{n-1})$.

1. For each $\mathbf{w} \in \{0,1\}^n$, compute $z_{\mathbf{w}} = \text{Dec}_{G_{f_{\mathbf{w}}}^{\text{plus}}}((m_{\mathbf{w},i})_{i \in [n]})$.
2. Output $z = \sum_{\mathbf{w} \in \{0,1\}^n} z_{\mathbf{w}}$.

Figure 5: A PSM protocol for F

for any $b \in \{0,1\}$ and $x_i \in X_i$. We set $\mathcal{F}^{\text{plus}} = \{f^{\text{plus}} : f \in \mathcal{F}\}$. We also define a function $f^{\text{affine}} : \{0,1\}^2 \times \prod_{i \in [n]} X_i \rightarrow \{0,1\}$ as

$$f^{\text{affine}}(a, b; (x_i)_{i \in [n]}) = af((x_i)_{i \in [n]}) + b \bmod 2$$

for any $(a, b) \in \{0,1\}^2$ and $x_i \in X_i$, and set $\mathcal{F}^{\text{affine}} = \{f^{\text{affine}} : f \in \mathcal{F}\}$. We assume that in a PSM protocol for f^{plus} or f^{affine} , the coefficients a, b are commonly known to all parties but are private and hidden from the referee. Finally, we define a function $H : \mathbb{F}_2^{n+2} \rightarrow \mathbb{F}_2$ as $H(z, b, (y_i)_{i \in [n]}) = z \prod_{i \in [n]} y_i + b$ for any $z, b, y_i \in \mathbb{F}_2$.

First, we reduce a PSM protocol for $F \in \mathcal{F}_{n,2d}$ to PSM protocols for G_f^{plus} with $f \in \mathcal{F}_{n,d}$.

Lemma 5.5. $\text{PSM}(\mathcal{F}_{n,2d}) \leq 2^n \cdot \max_{f \in \mathcal{F}_{n,d}} \text{PSM}(G_f^{\text{plus}})$.

Proof. Let $F \in \mathcal{F}_{n,2d}$. Suppose that for any $f \in \mathcal{F}_{n,d}$, we are given a PSM protocol for a function G_f^{plus} ,

$$\Pi_{G_f^{\text{plus}}} = (\text{Gen}_{G_f^{\text{plus}}}, \text{Enc}_{G_f^{\text{plus}}}, \text{Dec}_{G_f^{\text{plus}}}).$$

We show a PSM protocol Π_F for F in Figure 5.

First, we see the correctness of Π_F . Let $\mathbf{x} = (x_i)_{i \in [n]} \in [d]^n$, and $\mathbf{y} = (y_i)_{i \in [n]} \in \{0,1\}^n$. It follows from the correctness of $\Pi_{G_{f_{\mathbf{w}}}^{\text{plus}}}$ that

$$z_{\mathbf{w}} = G_{f_{\mathbf{w}}}^{\text{plus}}(b_{\mathbf{w}}; (x_i, \delta_{w_i}(y_i))_{i \in [n]}) = f_{\mathbf{w}}(\mathbf{x})\delta_{\mathbf{w}}(\mathbf{y}) + b_{\mathbf{w}}.$$

Since $\sum_{\mathbf{w} \in \{0,1\}^n} b_{\mathbf{w}} = 0$, it holds that

$$z = \sum_{\mathbf{w} \in \{0,1\}^n} f_{\mathbf{w}}(\mathbf{x})\delta_{\mathbf{w}}(\mathbf{y}) = f_{\mathbf{y}}(\mathbf{x}) = F((x_i, y_i)_{i \in [n]}).$$

Next, we see the privacy of Π_F . We have seen that $z_{\mathbf{w}} = f_{\mathbf{w}}(\mathbf{x})\delta_{\mathbf{w}}(\mathbf{y}) + b_{\mathbf{w}}$. Since $(b_{\mathbf{w}})_{\mathbf{w} \in \{0,1\}^n}$ are random additive shares of 0, $(z_{\mathbf{w}})_{\mathbf{w} \in \{0,1\}^n}$ are random additive shares of $z = \sum_{\mathbf{w} \in \{0,1\}^n} z_{\mathbf{w}} = F((x_i, y_i)_{i \in [n]})$. Thus,

Notations.

- Let $\Pi_{f^{\text{affine}}} = (\text{Gen}_{f^{\text{affine}}}, \text{Enc}_{f^{\text{affine}}}, \text{Dec}_{f^{\text{affine}}})$ be a PSM protocol for f^{affine} .
- Let $\hat{H} : \mathbb{F}_2^{n+2} \times \mathbb{F}_2^m \rightarrow \mathbb{F}_2^s$ be a DARE of H . That is,

$$\hat{H}((z, b, (y_i)_{i \in [n]}), \rho) = (h_0(z, \rho), h_1(b, \rho), (h'_i(y_i, \rho))_{i \in [n]})$$

for any $z, b, y_i \in \mathbb{F}_2$ and $\rho \in \mathbb{F}_2^m$. In particular, $h_0(z, \rho)$ can be written as $\kappa(\rho)z + \lambda(\rho)$ for some functions $\kappa, \lambda : \mathbb{F}_2^m \rightarrow \mathbb{F}_2^{s_0}$ with $s_0 \leq s$. Let $\text{Dec}_{\hat{H}}$ denote the decoding function of $\hat{H}$.

$\text{Gen}()$.

1. Choose $\rho \leftarrow_s \mathbb{F}_2^m$.
2. For each $j \in [s_0]$, let κ_j and λ_j be the j^{th} bit of $\kappa(\rho)$ and $\lambda(\rho)$, respectively.
3. For each $j \in [s_0]$, run $r_j \leftarrow \text{Gen}_{f^{\text{affine}}}()$.
4. Output $R = (\rho, (\kappa_j, \lambda_j, r_j)_{j \in [s_0]})$.

$\text{Enc}(i, (b, x_i), R)$.

1. If $i = 0$, compute $\hat{b} = h_1(b, \rho)$.
2. For each $j \in [s_0]$, compute $m_{i,j} = \text{Enc}_{f^{\text{affine}}}(i, (\kappa_j, \lambda_j, x_i), r_j)$.
3. Set $\mu_i = (m_{i,j})_{j \in [s_0]}$.
4. Compute $\hat{y}_i = h'_i(y_i, \rho)$.
5. Output $M_i = (\mu_i, \hat{b}, \hat{y}_i)$.

$\text{Dec}(M_0, \dots, M_{n-1})$.

1. Parse μ_i into $(m_{i,j})_{j \in [s_0]}$.
2. For each $j \in [s_0]$, compute $\hat{z}_j = \text{Dec}_{f^{\text{affine}}}((m_{i,j})_{i \in [n]})$.
3. Set $\hat{z} = (\hat{z}_j)_{j \in [s_0]}$.
4. Output $w = \text{Dec}_{\hat{H}}(\hat{z}, \hat{b}, (\hat{y}_i)_{i \in [n]})$.

Figure 6: A PSM protocol for G_f^{plus}

we have a randomized algorithm Sim_0 that simulates the distribution of $(z_w)_{w \in \{0,1\}^n}$ from z . Furthermore, for each $w \in \{0,1\}^n$, the privacy of $\Pi_{G_{f_w}^{\text{plus}}}$ ensures the existence of a simulator $\text{Sim}_{G_{f_w}^{\text{plus}}}$ that simulates the distribution of $(m_{w,i})_{i \in [n]}$ from z_w . Therefore, since all $\Pi_{G_{f_w}^{\text{plus}}}$'s are executed on independent randomness, we obtain a simulator that simulates the distribution of messages $(m_{w,i})_{w \in \{0,1\}^n, i \in [n]}$ in Π_F from z .

Since $f_w \in \mathcal{F}_{n,d}$, the communication complexity of each $\Pi_{G_{f_w}^{\text{plus}}}$ is upper bounded by $\max_{f \in \mathcal{F}_{n,d}} \text{PSM}(G_f^{\text{plus}})$. Since Π_F internally executes 2^n such protocols, the communication complexity of Π_F is upper bounded by $2^n \cdot \max_{f \in \mathcal{F}_{n,d}} \text{PSM}(G_f^{\text{plus}})$. $\square$

Next, we reduce a PSM protocol for G_f^{plus} to one for f^{affine} .

Lemma 5.6. *For any $f \in \mathcal{F}_{n,d}$, $\text{PSM}(G_f^{\text{plus}}) \leq s \cdot \text{PSM}(f^{\text{affine}}) \leq s \cdot \text{PSM}(\mathcal{F}_{n,d}^{\text{affine}})$, where s is the output length of any DARE of H .*

Proof. Let $f \in \mathcal{F}_{n,d}$ and suppose that a PSM protocol for f^{affine} is given. We show a PSM protocol $\Pi_{G_f^{\text{plus}}}$ for G_f^{plus} in Figure 6.

Notations.

- Suppose that f is not the constant function that always outputs 1. Let $(x_i^*)_{i \in [n]}$ be an input such that $f((x_i^*)_{i \in [n]}) = 0$.
- Let $\Pi_{f^{\text{plus}}} = (\text{Gen}_{f^{\text{plus}}}, \text{Enc}_{f^{\text{plus}}}, \text{Dec}_{f^{\text{plus}}})$ be a PSM protocol for f^{plus} .

$\text{Gen}()$. Output $r \leftarrow \text{Gen}_{f^{\text{plus}}}()$.

$\text{Enc}(i, (a, b, x_i), r)$.

1. If $a = 0$, output $m_i = \text{Enc}_{f^{\text{plus}}}(i, (b, x_i^*), r)$.
2. If $a = 1$, output $m_i = \text{Enc}_{f^{\text{plus}}}(i, (b, x_i), r)$.

$\text{Dec}(m_0, \dots, m_{n-1})$. Output $y = \text{Dec}_{f^{\text{plus}}}(m_0, \dots, m_{n-1})$.

Figure 7: A PSM protocol for f^{affine}

First, we see the correctness of $\Pi_{G_f^{\text{plus}}}$. Let $b \in \{0, 1\}$, $\mathbf{x} = (x_i)_{i \in [n]} \in [d]^n$, and $\mathbf{y} = (y_i)_{i \in [n]} \in \{0, 1\}^n$. It follows from the correctness of $\Pi_{f^{\text{affine}}}$ that

$$\hat{z}_j = f^{\text{affine}}(\kappa_j, \lambda_j; (x_i)_{i \in [n]}) = \kappa_j f(\mathbf{x}) + \lambda_j$$

for each $j \in [s_0]$, and hence that $\hat{z} = h_0(f(\mathbf{x}), \rho)$. Therefore, we have that

$$w = H(f(\mathbf{x}), b, \mathbf{y}) = f(\mathbf{x}) \prod_{i \in [n]} y_i + b = G_f^{\text{plus}}(b; (x_i, y_i)_{i \in [n]}).$$

Next, we see the privacy of $\Pi_{G_f^{\text{plus}}}$. The privacy of the DARE $\hat{H}$ ensures the existence of a simulator $\text{Sim}_{\hat{H}}$ that simulates the distribution of $(\hat{z}, \hat{b}, (\hat{y}_i)_{i \in [n]})$ from $w = H(f(\mathbf{x}), b, \mathbf{y})$. Furthermore, the privacy of $\Pi_{f^{\text{affine}}}$ ensures the existence of a simulator $\text{Sim}_{f^{\text{affine}}}$ that simulates the distribution of $(\mu_i)_{i \in [n]}$ from $\hat{z}$. Therefore, we have a simulator that simulates the distribution of messages $((\mu_i)_{i \in [n]}, \hat{b}, (\hat{y}_i)_{i \in [n]})$ in $\Pi_{G_f^{\text{plus}}}$ from w .

We have that $\sum_{i \in [n]} |\mu_i| \leq s_0 \cdot \text{PSM}(f^{\text{affine}})$. Furthermore, $|\hat{b}| + \sum_{i \in [n]} |\hat{y}_i| \leq s - s_0 \leq (s - s_0) \cdot \text{PSM}(f^{\text{affine}})$. Thus, the communication complexity of $\Pi_{G_f^{\text{plus}}}$ is at most $s \cdot \text{PSM}(f^{\text{affine}})$. $\square$

Since H is computed by an arithmetic formula of depth at most $\log(n+2)$, it follows from Proposition 3.5 that $s = O(n^{1+\epsilon})$ for any constant $\epsilon > 0$.

Next, we reduce a PSM protocol for f^{affine} to one for f^{plus} .

Lemma 5.7. *For any $f : \prod_{i \in [n]} X_i \rightarrow \{0, 1\}$, $\text{PSM}(f^{\text{affine}}) \leq \text{PSM}(f^{\text{plus}})$. In particular, $\text{PSM}(\mathcal{F}_{n,d}^{\text{affine}}) \leq \text{PSM}(\mathcal{F}_{n,d}^{\text{plus}})$.*

Proof. If f is the constant function that always outputs 1, then $\text{PSM}(f^{\text{affine}}) = 1 \leq \text{PSM}(f^{\text{plus}})$ since the referee can learn $f^{\text{affine}}(a, b; (x_i)_{i \in [n]})$ by letting the first party send the bit $a + b$.

Suppose that f is not the constant function 1. Then, there exists an input $(x_i^*)_{i \in [n]}$ such that $f((x_i^*)_{i \in [n]}) = 0$. We fix $(x_i^*)_{i \in [n]}$. In Figure 7, we construct a PSM protocol $\Pi_{f^{\text{affine}}}$ for f^{affine} from any given PSM protocol for f^{plus} .

We see the correctness of $\Pi_{f^{\text{affine}}}$. Let $(a, b) \in \{0, 1\}^2$ and $\mathbf{x} = (x_i)_{i \in [n]} \in \prod_{i \in [n]} X_i$. If $a = 0$, then the correctness of $\Pi_{f^{\text{plus}}}$ ensures that

$$y = f^{\text{plus}}(b; (x_i^*)_{i \in [n]}) = f((x_i^*)_{i \in [n]}) + b = b = f^{\text{affine}}(0, b; (x_i)_{i \in [n]}).$$

If $a = 1$, then we have that

$$y = f^{\text{plus}}(b; (x_i)_{i \in [n]}) = f((x_i)_{i \in [n]}) + b = f^{\text{affine}}(1, b; (x_i)_{i \in [n]}).$$

The privacy of $\Pi_{f^{\text{affine}}}$ is straightforward since the privacy of $\Pi_{f^{\text{plus}}}$ ensures that the messages are simulated from the function f and the output $y = f^{\text{affine}}(a, b; (x_i)_{i \in [n]})$. Clearly, the communication complexity of $\Pi_{f^{\text{affine}}}$ is at most that of $\Pi_{f^{\text{plus}}}$. $\square$

Finally, we reduce a PSM protocol for f^{plus} to one for $f' \in \mathcal{F}_{n, 2\lceil d/2 \rceil}$. Note that $\mathcal{F}_{n, 2\lceil d/2 \rceil} = \mathcal{F}_{n, d}$ when d is even.

Lemma 5.8. $\text{PSM}(\mathcal{F}_{n, d}^{\text{plus}}) \leq O(1) \cdot \text{PSM}(\mathcal{F}_{n, 2\lceil d/2 \rceil})$.

Proof. Let $f^{\text{plus}} \in \mathcal{F}_{n, d}^{\text{plus}}$. We define $g : [2d] \times [d]^{n-1} \rightarrow \{0, 1\}$ as

$$g((x_0, b), x_1, \dots, x_{n-1}) = f^{\text{plus}}(b; x_0, x_1, \dots, x_{n-1})$$

for any $x_i \in [d]$ and $b \in \{0, 1\}$, where we fix a one-to-one correspondence between $x' \in [2d]$ and $(x, b) \in [d] \times \{0, 1\}$. Thus, a PSM protocol for g can be used as a PSM protocol for f^{plus} since party 0 knows b . We obtain $\text{PSM}(\mathcal{F}_{n, d}^{\text{plus}}) \leq \text{PSM}(\mathcal{F}_{2d, d, \dots, d})$.

Let $g \in \mathcal{F}_{(2d, d, \dots, d)}$. There exists an injective map from $[2d]$ to $[e] \times \{0, 1\}^2$, where $e = \lceil d/2 \rceil$, and hence each element $x_0 \in [2d]$ is uniquely represented as $(x'_0, \mathbf{u})$, where $x'_0 \in [e]$ and $\mathbf{u} \in \{0, 1\}^2$. For each $\mathbf{w} \in \{0, 1\}^2$, we define $g_{\mathbf{w}} \in \mathcal{F}_{(e, d, \dots, d)}$ as $g_{\mathbf{w}}(x'_0, x_1, \dots, x_{n-1}) = g((x'_0, \mathbf{w}), x_1, \dots, x_{n-1})$ for any $x'_0 \in [e]$ and $x_i \in [d]$. Below, we will show that $\text{PSM}(g) \leq O(1) \cdot \max_{\mathbf{w} \in \{0, 1\}^2} \text{PSM}(g_{\mathbf{w}}^{\text{affine}})$. Note that $\text{PSM}(g_{\mathbf{w}}^{\text{affine}}) \leq \text{PSM}(g_{\mathbf{w}}^{\text{plus}})$ from Lemma 5.7 and $\text{PSM}(g_{\mathbf{w}}^{\text{plus}}) \leq \text{PSM}(\mathcal{F}_{(2e, d, \dots, d)})$ from a similar argument as above. We then obtain $\text{PSM}(g) \leq O(1) \cdot \text{PSM}(\mathcal{F}_{(2e, d, \dots, d)})$. In summary, we conclude that

$$\text{PSM}(\mathcal{F}_{n, d}^{\text{plus}}) \leq \text{PSM}(\mathcal{F}_{(2d, d, \dots, d)}) \leq O(1) \cdot \text{PSM}(\mathcal{F}_{(e, d, \dots, d)}^{\text{plus}}) \leq O(1) \cdot \text{PSM}(\mathcal{F}_{(2e, d, \dots, d)}) \leq O(1) \cdot \text{PSM}(\mathcal{F}_{n, 2e}),$$

which completes the proof.

We show that $\text{PSM}(g) \leq O(1) \cdot \max_{\mathbf{w} \in \{0, 1\}^2} \text{PSM}(g_{\mathbf{w}}^{\text{affine}})$. First, we observe that

$$g((x'_0, \mathbf{u}), x_1, \dots, x_{n-1}) = \sum_{\mathbf{w} \in \{0, 1\}^2} g_{\mathbf{w}}(x'_0, x_1, \dots, x_{n-1}) \cdot \delta_{\mathbf{w}}(\mathbf{u}),$$

where $\delta_{\mathbf{w}}(\mathbf{u})$ is 1 if $\mathbf{u} = \mathbf{w}$ and 0 otherwise. We define a function $K : \mathbb{F}_2^8 \rightarrow \mathbb{F}_2$ as $K((z_{\mathbf{w}})_{\mathbf{w} \in \{0, 1\}^2}, (\delta_{\mathbf{w}})_{\mathbf{w} \in \{0, 1\}^2}) = \sum_{\mathbf{w} \in \{0, 1\}^2} z_{\mathbf{w}} \delta_{\mathbf{w}}$ for any $z_{\mathbf{w}}, \delta_{\mathbf{w}} \in \mathbb{F}_2$. Suppose that for each $\mathbf{w} \in \{0, 1\}^2$, a PSM protocol $\Pi_{g_{\mathbf{w}}^{\text{affine}}}$ for $g_{\mathbf{w}}^{\text{affine}}$ is given and that a DARE $\hat{K}$ of K is also given. We construct a PSM protocol Π_g for g in Figure 8.

First, we see the correctness of Π_g . Let $x_0 = (x'_0, \mathbf{u}) \in [e] \times \{0, 1\}^2$ and $x_1, \dots, x_{n-1} \in [d]$. It follows from the correctness of $\Pi_{g_{\mathbf{w}}^{\text{affine}}}$ that

$$\hat{z}_{\mathbf{w}, j} = g_{\mathbf{w}}^{\text{affine}}(\kappa_{\mathbf{w}, j}, \lambda_{\mathbf{w}, j}; x'_0, x_1, \dots, x_{n-1}) = \kappa_{\mathbf{w}, j} \cdot g_{\mathbf{w}}(x'_0, x_1, \dots, x_{n-1}) + \lambda_{\mathbf{w}, j}$$

for each $j \in [s_0]$, and hence that $\hat{z}_{\mathbf{w}} = k_{\mathbf{w}}(z_{\mathbf{w}}, \rho)$ where $z_{\mathbf{w}} = g_{\mathbf{w}}(x'_0, x_1, \dots, x_{n-1})$. Therefore, we have that

$$y = K((z_{\mathbf{w}})_{\mathbf{w} \in \{0, 1\}^2}, (\delta_{\mathbf{w}}(\mathbf{u}))_{\mathbf{w} \in \{0, 1\}^2}) = z_{\mathbf{u}} = g((x'_0, \mathbf{u}), x_1, \dots, x_{n-1}).$$

Next, we see the privacy of Π_g . The privacy of the DARE $\hat{K}$ ensures the existence of a simulator $\text{Sim}_{\hat{K}}$ that simulates the distribution of $((\hat{z}_{\mathbf{w}})_{\mathbf{w} \in \{0, 1\}^2}, (\hat{\delta}_{\mathbf{w}})_{\mathbf{w} \in \{0, 1\}^2})$ from $y = K((z_{\mathbf{w}})_{\mathbf{w} \in \{0, 1\}^2}, (\delta_{\mathbf{w}}(\mathbf{u}))_{\mathbf{w} \in \{0, 1\}^2})$. Furthermore, the privacy of $\Pi_{g_{\mathbf{w}}^{\text{affine}}}$ ensures the existence of a simulator $\text{Sim}_{g_{\mathbf{w}}^{\text{affine}}}$ that simulates the distribution of $(\mu_{i, \mathbf{w}})_{i \in [n]}$ from $\hat{z}_{\mathbf{w}}$. Therefore, we have a simulator that simulates the distribution of messages $((\mu_{i, \mathbf{w}})_{i \in [n], \mathbf{w} \in \{0, 1\}^2}, (\hat{\delta}_{\mathbf{w}})_{\mathbf{w} \in \{0, 1\}^2})$ in Π_g from y .

We have that $\sum_{i \in [n]} |\mu_{i, \mathbf{w}}| \leq t_0 \cdot \text{PSM}(g_{\mathbf{w}}^{\text{affine}})$ for each $\mathbf{w}$. Furthermore, $|\hat{\delta}_{\mathbf{w}}| \leq t - t_0$. Note that since K is a function of constant size, we have that $t = O(1)$. Thus, the communication complexity of Π_g is at most $\sum_{\mathbf{w} \in \{0, 1\}^2} t \cdot \text{PSM}(g_{\mathbf{w}}^{\text{affine}}) = O(1) \cdot \max_{\mathbf{w} \in \{0, 1\}^2} \text{PSM}(g_{\mathbf{w}}^{\text{affine}})$. $\square$

Notations.

- Let $\Pi_{g_{\mathbf{w}}^{\text{affine}}} = (\text{Gen}_{g_{\mathbf{w}}^{\text{affine}}}, \text{Enc}_{g_{\mathbf{w}}^{\text{affine}}}, \text{Dec}_{g_{\mathbf{w}}^{\text{affine}}})$ be a PSM protocol for $g_{\mathbf{w}}^{\text{affine}}$.
- Let $\hat{K} : \mathbb{F}_2^8 \times \mathbb{F}_2^m \rightarrow \mathbb{F}_2^t$ be a DARE of K . That is,

$$\hat{K}((z_{\mathbf{w}})_{\mathbf{w} \in \{0,1\}^2}, (\delta_{\mathbf{w}})_{\mathbf{w} \in \{0,1\}^2}, \rho) = ((k_{\mathbf{w}}(z_{\mathbf{w}}, \rho))_{\mathbf{w} \in \{0,1\}^2}, (k'_{\mathbf{w}}(\delta_{\mathbf{w}}, \rho))_{\mathbf{w} \in \{0,1\}^2})$$

for any $z_{\mathbf{w}}, \delta_{\mathbf{w}} \in \mathbb{F}_2$ and $\rho \in \mathbb{F}_2^m$. In particular, $k_{\mathbf{w}}(z_{\mathbf{w}}, \rho)$ can be written as $\kappa_{\mathbf{w}}(\rho)z_{\mathbf{w}} + \lambda_{\mathbf{w}}(\rho)$ for some functions $\kappa_{\mathbf{w}}, \lambda_{\mathbf{w}} : \mathbb{F}_2^m \rightarrow \mathbb{F}_2^{t_0}$ with $t_0 \leq t$. Let $\text{Dec}_{\hat{K}}$ denote the decoding function of $\hat{K}$.

$\text{Gen}()$.

1. Choose $\rho \leftarrow_{\$} \mathbb{F}_2^m$.
2. For each $\mathbf{w} \in \{0,1\}^2$ and $j \in [t_0]$, let $\kappa_{\mathbf{w},j}$ and $\lambda_{\mathbf{w},j}$ be the j^{th} bit of $\kappa_{\mathbf{w}}(\rho)$ and $\lambda_{\mathbf{w}}(\rho)$, respectively.
3. For each $\mathbf{w} \in \{0,1\}^2$ and $j \in [t_0]$, run $r_{\mathbf{w},j} \leftarrow \text{Gen}_{g_{\mathbf{w}}^{\text{affine}}}()$.
4. Output $R = (\rho, (\kappa_{\mathbf{w},j}, \lambda_{\mathbf{w},j}, r_{\mathbf{w},j})_{\mathbf{w} \in \{0,1\}^2, j \in [t_0]})$.

$\text{Enc}(i, x_i, R)$.

- If $i = 0$:
 1. Parse $x_0 = (x'_0, \mathbf{u})$ where $x'_0 \in [e]$ and $\mathbf{u} \in \{0,1\}^2$.
 2. For each $\mathbf{w} \in \{0,1\}^2$:
 - (a) For each $j \in [t_0]$, compute $m_{0,\mathbf{w},j} = \text{Enc}_{g_{\mathbf{w}}^{\text{affine}}}(0, (\kappa_{\mathbf{w},j}, \lambda_{\mathbf{w},j}, x'_0), r_{\mathbf{w},j})$.
 - (b) Set $\mu_{0,\mathbf{w}} = (m_{0,\mathbf{w},j})_{j \in [t_0]}$.
 - (c) Compute $\hat{\delta}_{\mathbf{w}} = k'_{\mathbf{w}}(\delta_{\mathbf{w}}(\mathbf{u}), \rho)$.
 3. Output $M_0 = ((\mu_{0,\mathbf{w}})_{\mathbf{w} \in \{0,1\}^2}, (\hat{\delta}_{\mathbf{w}})_{\mathbf{w} \in \{0,1\}^2})$.
- If $i \neq 0$:
 1. For each $\mathbf{w} \in \{0,1\}^2$:
 - (a) For each $j \in [t_0]$, compute $m_{i,\mathbf{w},j} = \text{Enc}_{g_{\mathbf{w}}^{\text{affine}}}(i, (\kappa_{\mathbf{w},j}, \lambda_{\mathbf{w},j}, x'_0), r_{\mathbf{w},j})$.
 - (b) Set $\mu_{i,\mathbf{w}} = (m_{i,\mathbf{w},j})_{j \in [t_0]}$.
 2. Output $M_i = ((\mu_{i,\mathbf{w}})_{\mathbf{w} \in \{0,1\}^2})$.

$\text{Dec}(M_0, \dots, M_{n-1})$.

1. For each $\mathbf{w} \in \{0,1\}^2$:
 - (a) Parse each $\mu_{i,\mathbf{w}}$ into $(m_{i,\mathbf{w},j})_{j \in [s_0]}$.
 - (b) For each $j \in [s_0]$, compute $\hat{z}_{\mathbf{w},j} = \text{Dec}_{g_{\mathbf{w}}^{\text{affine}}}((m_{i,\mathbf{w},j})_{i \in [n]})$.
 - (c) Set $\hat{z}_{\mathbf{w}} = (\hat{z}_{\mathbf{w},j})_{j \in [s_0]}$.
2. Output $y = \text{Dec}_{\hat{K}}((\hat{z}_{\mathbf{w}})_{\mathbf{w} \in \{0,1\}^2}, (\hat{\delta}_{\mathbf{w}})_{\mathbf{w} \in \{0,1\}^2})$.

Figure 8: A PSM protocol for g

Lemmas 5.5, 5.6, 5.7, and 5.8 yield the following inequality.

Lemma 5.9. *For a parameter $c_n = O(2^n n^{1+\epsilon})$ where $\epsilon > 0$ is any constant,*

$$\text{PSM}(\mathcal{F}_{n,2d}) \leq c_n \cdot \text{PSM}(\mathcal{F}_{n,2\lceil d/2 \rceil}).$$

By applying Lemma 5.9 iteratively, we obtain the following proposition.

Proposition 5.10. *For $d_1 \leq d_2$,*

$$\text{PSM}(\mathcal{F}_{n,d_2}) \leq (O(d_2/d_1))^{n+O(\log n)} \cdot \text{PSM}(\mathcal{F}_{n,d_1}).$$

Proof. Let ℓ_1 be the largest integer such that $2^{\ell_1} \leq d_1$. Let ℓ_2 be the smallest integer such that $d_2 \leq 2^{\ell_2}$. It then holds that

$$\text{PSM}(\mathcal{F}_{n,d_2}) \leq \text{PSM}(\mathcal{F}_{n,2^{\ell_2}}) \leq \text{PSM}(\mathcal{F}_{n,2^{\ell_2-1}}) \cdot c_n \leq \dots \leq \text{PSM}(\mathcal{F}_{n,2^{\ell_1}}) \cdot c_n^{\ell_2-\ell_1} \leq \text{PSM}(\mathcal{F}_{n,d_1}) \cdot c_n^{\ell_2-\ell_1}.$$

Since $\ell_2 - \ell_1 \leq \log(d_2/d_1) + 2$ and $c_n = O(2^n n^{1+\epsilon})$, the stated inequality is obtained. $\square$

Now, we give a proof of Theorem 1.2 in the case of $\mathfrak{C} = \text{PSM}$. From Theorem 1.1, we have an inequality $\text{PSM}(\mathcal{S}_{n,d}) \geq \text{PSM}(\mathcal{F}_{n,d'})$ for $d' = \lfloor d/n \rfloor$. Since $d/d' = O(n)$, it follows from Proposition 5.10 that

$$\text{PSM}(\mathcal{F}_{n,d'}) \geq n^{-O(n)} \cdot \text{PSM}(\mathcal{F}_{n,d}),$$

which proves the lower bound in (3). On the other hand, we have an inequality $\text{PSM}(\mathcal{S}_{n,d}) \geq \text{PSM}(\mathcal{F}_{d-1,n'+1})$ for $n' = \lfloor n/(d-1) \rfloor$. Since $n/n' = O(d)$, it similarly follows from Proposition 5.10 that

$$\text{PSM}(\mathcal{F}_{d-1,n'+1}) \geq d^{-O(d)} \cdot \text{PSM}(\mathcal{F}_{d-1,n+1}),$$

which proves the lower bound in (4).

Remark 5.11. We discuss the tightness of our bounds on $\text{PSM}(\mathcal{S}_{n,d})$ and $\text{CDS}(\mathcal{S}_{n,d})$. The bounds (3) in Theorem 1.2 imply that for $\mathfrak{C} \in \{\text{PSM}, \text{CDS}\}$, $\mathfrak{C}(\mathcal{S}_{n,d})$ is equal to $\mathfrak{C}(\mathcal{F}_{n,d})$ up to a factor of $n^{O(n)}$. This shows the tightness up to a constant when $n = O(1)$. It is important to note that for PSM and CDS protocols with a constant number of parties, the efficiency gain achievable by focusing on the subclass $\mathcal{S}_{n,d}$ is limited to a constant-factor improvement. On the other hand, the bounds (4) imply that $\mathfrak{C}(\mathcal{S}_{n,d})$ is determined by $\mathfrak{C}(\mathcal{F}_{d-1,n+1})$ up to a multiplicative factor of $n^2 d^{O(d)}$. In the PSM case, since the best known bound for $\text{PSM}(\mathcal{F}_{d-1,n+1})$ is dominated by $n^{d/2}$, this remaining factor is asymptotically smaller when $d \ll n$, e.g., $d = 2^{o(\log n)}$. In the CDS case, if $d = \Theta(\log n / \log \log n)$, this remaining factor is polynomial in n and is asymptotically smaller than the dominant term $2^{O(\sqrt{d} \log n \log(d \log n))} = n^{\omega(1)}$ in the best known bound for $\text{CDS}(\mathcal{F}_{d-1,n+1})$ [36].

6 Applications to Related Models

Our new constructions of PSM and CDS for symmetric functions lead to improvements over the state-of-the-art results in related models. Our PSM protocol can be extended to ad hoc PSM [11, 14] and it improves the best known complexity for symmetric functions [2]. Our CDS protocol implies a new efficient secret sharing scheme for multipartite uniform access structures.

6.1 Ad hoc PSM for Symmetric Functions

A k -out-of- n ad hoc PSM protocol [14] is an advanced notion of an n -party PSM protocol. In this model, a subset S of k input parties, where S is not known in advance, actually send messages and the referee is able to evaluate a k -input function f on the subset of inputs $(x_i)_{i \in S}$.

Definition 6.1. *Let X and Y be finite sets. Let $f : X^k \rightarrow Y$ be a function. A k -out-of- n ad hoc PSM protocol Π for f is a tuple of algorithms $(\text{Gen}, \text{Enc}, \text{Dec})$, where:*

- $\text{Gen}() \rightarrow r$: Gen is a randomized algorithm that takes no input and outputs common randomness $r \in \{0, 1\}^{\text{rand}}$.
- $\text{Enc}(i, x_i, r) \rightarrow m_i$: Enc is a deterministic algorithm that takes an index i , $x_i \in X$, and $r \in \{0, 1\}^{\text{rand}}$ as input and outputs a message $m_i \in \{0, 1\}^{\text{comm}_i}$.
- $\text{Dec}(S, (m_i)_{i \in S}) \rightarrow y$: Dec is a deterministic algorithm that takes a k -sized subset $S \subseteq [n]$ and k messages $(m_i)_{i \in S}$ as input and outputs $y \in Y$.

satisfying the following properties:

Correctness. For any k -sized subset $S \subseteq [n]$ and any k inputs $x_S = (x_i)_{i \in S} \in X^k$, it holds that

$$\text{Dec}(S, (\text{Enc}(i, f(x_i), r))_{i \in S}) = f(x_S),$$

with probability 1 over the choice of $r \leftarrow \text{Gen}()$.

Privacy.

- There exists a randomized algorithm Sim such that for any k -sized subset $S \subseteq [n]$ and any k inputs $x_S \in X^k$, it holds that

$$(\text{Enc}(i, x_i, r))_{i \in S} \equiv \text{Sim}(S, f(x_S)),$$

where the probabilities are taken over $r \leftarrow \text{Gen}()$ and the random coins of Sim .

- There exists a randomized algorithm Sim such that for any $k' < k$, any k' -sized subset $S \subseteq [n]$ and any k' inputs $x_S \in X^{k'}$, it holds that

$$(\text{Enc}(i, x_i, r))_{i \in S} \equiv \text{Sim}(S),$$

where the probabilities are taken over $r \leftarrow \text{Gen}()$ and the random coins of Sim .

It is natural in this ad-hoc setting to assume that f is a symmetric function independent of the identities of the parties. There exist incomparable compilers from a k -party PSM protocol for $f \in \mathcal{S}_{k,d}$ to an ad hoc PSM protocol for f [2, 19]. The resulting communication complexity are $O(kn) \cdot \text{PSM}(f)$ [2, 19] or $O(k^5 \log n) \cdot \text{PSM}(f)$. However, the symmetry of f was not exploited and the trivial bound $\text{PSM}(f) \leq \text{PSM}(\mathcal{F}_{n,d})$ was applied in [2]. Consequently, the previous upper bounds suffer from an exponential factor of $d^{n/2}$. On the other hand, if our improved upper bound on symmetric functions is applied, then we obtain an ad hoc PSM protocol Π for $f \in \mathcal{S}_{k,d}$ such that

- $\text{Comm}(\Pi) = O(k^3 n) \cdot \text{PSM}(\mathcal{F}_{d-1,k+1}) = O((k+1)^{\frac{d+5}{2}} d^3 n)$; or
- $\text{Comm}(\Pi) = O(k^7 \log n) \cdot \text{PSM}(\mathcal{F}_{d-1,k+1}) = O((k+1)^{\frac{d+13}{2}} d^3 \log n)$.

In addition to this immediate application, we propose a direct construction of ad hoc PSM for symmetric functions without using the generic compiler of [2, 19]. This achieves communication complexity $O((k+1)^{\frac{d+3}{2}} d^3 n)$, further improving the first upper bound mentioned above by a factor of k .

Recall that in our PSM protocol for symmetric functions, we build a histogram in which each bin has the value $\sum_{i \in [n]} \delta_{j,i}$, where $\delta_{j,i} = 1$ if $x_i = j$ and 0 otherwise. Then, the parties run a PSM protocol for the selection function Sel_n to reveal only the message stored in the entry corresponding to $\sum_{i \in [n]} \delta_{j,i}$. In the ad-hoc setting where a set S of k parties participate, we have to reveal the message corresponding to $\sum_{i: P_i \in S} \delta_{j,i}$. This issue can be easily addressed by a method of sharing dummy messages for $P_i \notin S$, inspired by [14, 24]: we distribute a message that $P_i \notin S$ would compute on input $\delta_{j,i} = 0$ in the protocol for Sel_n , using a k -out-of- n secret sharing scheme. This ensures that the missing $n - k$ messages are filled with the dummy messages so that the referee can proceed to the decoding phase. However, since the protocol for Sel_n requires $O(n^2)$ communication, this results in $O(kn^2) \cdot \text{PSM}(\mathcal{F}_{d-1,k+1})$, which does not yield improvement. To address this issue, we observe that the value $\sum_{i \in [n]} \delta_{j,i}$ of each bin is always at most k in the ad-hoc setting, since only k parties contribute to the histogram. Consequently, we can reduce an ad hoc PSM protocol to

a *restricted version* of the selection function, $\text{Sel}_{k,n}$. Specifically, define a subset $S \subseteq \{0,1\}^{k+1} \times \{0,1\}^n$ as $S = \{(\mathbf{m}, (x_i)_{i \in [n]}) : \sum_{i \in [n]} x_i \leq k\}$. Define a partial function $\text{Sel}_{k,n} : \{0,1\}^{k+1} \times \{0,1\}^n \rightarrow \{0,1,\star\}$ defined over S as $\text{Sel}_{k,n}(\mathbf{m}; (x_i)_{i \in [n]}) = \mathbf{m}[s]$ for any $(\mathbf{m}, (x_i)_{i \in [n]}) \in S$, where $s = \sum_{i \in [n]} x_i \leq k$ and $\mathbf{m}[s]$ is the s^{th} element of $\mathbf{m}$. By exploiting the reduced complexity of $\text{Sel}_{k,n}$, we construct a PSM protocol with improved communication complexity $O(kn)$ (see Appendix B for details). The overall communication complexity of our ad hoc protocol for symmetric functions becomes $O(k^2n) \cdot \text{PSM}(\mathcal{F}_{d-1,k+1})$. The formal proof is provided in Appendix C.

Theorem 6.2. *For any $f \in \mathcal{S}_{k,d}$, there exists a k -out-of- n ad hoc PSM protocol Π for f such that $\text{Comm}(\Pi) = O(k^2n) \cdot \text{PSM}(\mathcal{F}_{d-1,k+1}) + O(kd \log n)$. In particular, it holds that $O((k+1)^{\frac{d+3}{2}} d^3 n)$.*

6.2 Secret Sharing for Multipartite Uniform Access Structures

Let $\{P_0, \dots, P_{n-1}\}$ the set of n parties. We say that a collection Γ of subsets of parties is an access structure if it is monotonically increasing, that is, $A \in \Gamma$ implies $B \in \Gamma$ for any $A \subseteq B$. A secret sharing scheme for an access structure Γ divides a secret s into n shares $(v_i)_{i \in [n]}$ such that a subset of shares $(v_i)_{i \in A}$ can recover s if $A \in \Gamma$ and reveals nothing if $A \notin \Gamma$.

Definition 6.3. *Let Γ be an access structure on $[n]$. A secret sharing scheme Π for Γ is a tuple of algorithms $(\text{Share}, \text{Rec})$, where:*

- $\text{Share}(s) \rightarrow (v_1, \dots, v_n)$: *Share is a randomized algorithm that takes a secret $s \in \{0,1\}$ as input and outputs shares $(v_i)_{i \in [n]} \in \prod_{i \in [n]} \{0,1\}^{\text{len}_i}$.*
- $\text{Rec}(A, (v_i)_{i \in A}) \rightarrow s'$: *Rec is a deterministic algorithm that takes a set A and shares $v_i \in \{0,1\}^{\text{len}_i}$ for all $i \in A$ as input and outputs $s' \in \{0,1\}$.*

satisfying the following properties:

Correctness. *For any $s \in \{0,1\}$ and any $A \in \Gamma$, it holds that*

$$\Pr[(v_i)_{i \in [n]} \leftarrow \text{Share}(s) : \text{Rec}(A, (v_i)_{i \in A}) = s] = 1.$$

Privacy. *For any $A \notin \Gamma$, it holds that*

$$\text{Share}(0)_A \equiv \text{Share}(1)_A,$$

where $\text{Share}(s)_A = (v_i)_{i \in A}$ for $(v_i)_{i \in [n]} \leftarrow \text{Share}(s)$.

The (total) share size of Π is defined as $\text{ss}(\Pi) = \sum_{i \in [n]} \text{len}_i$.

We say that an access structure Γ is k -uniform if $A \in \Gamma$ for any set A with $|A| > k$ and $A \notin \Gamma$ for any set A with $|A| < k$. Thus, Γ uniquely corresponds to a function $g_\Gamma : [n]^k \rightarrow \{0,1\}$ defined as $g_\Gamma(i_0, \dots, i_{k-1}) = 1$ if and only if $A = \{P_{i_0}, \dots, P_{i_{k-1}}\} \in \Gamma$. It was shown in [2] that a k -party CDS protocol Π for g_Γ implies a secret sharing scheme for Γ whose share size is approximately bounded above by $\text{CDS}(g_\Gamma)$. However, while $g_\Gamma \in \mathcal{S}_{k,n}$, exploiting its symmetry cannot yield an improvement since $k < n$. Consequently, the share size for Γ incurs a sub-exponential factor $nk2^{\tilde{O}(\sqrt{k \log n})}$ in k .

We show that our construction of CDS specialized for symmetric functions is effective for a special class of *multipartite* uniform access structures. We say that an access structure Γ is d -partite if there exists a partition $(S_i)_{i \in [d]}$ of the set of n parties such that $A \in \Gamma$ implies $\sigma(A) \in \Gamma$ for any permutation σ that preserves the partition, i.e., $\sigma(S_i) = S_i$ for all i . In particular, as Γ is monotonically increasing, it is uniquely determined by the set of *minimal* vectors in $\mathcal{P}_\Gamma = \{(|A \cap S_i|)_{i \in [d]} : A \in \Gamma\}$ (with respect to the lexicographical order). Our key observation is that a d -partite k -uniform access structure Γ uniquely corresponds to a function $h_\Gamma : [d]^k \rightarrow \{0,1\}$ defined as

$$A = \{P_{i_0}, \dots, P_{i_{k-1}}\} \in \Gamma \iff h_\Gamma(j_0, \dots, j_{k-1}) = 1 \quad (9)$$

where each $j_h \in [d]$ denotes the part to which P_{i_h} belongs, i.e., $i_h \in S_{j_h}$. Note that $h_\Gamma \in \mathcal{S}_{k,d}$. We show a transformation from a CDS protocol for h_Γ to a secret sharing scheme for a d -partite k -uniform access structure Γ . Since $h_\Gamma = g_\Gamma$ when $d = n$, this generalizes the previous transformation [2]. The formal proof is given in Appendix D.

Theorem 6.4. *Let Γ be a d -partite k -uniform access structure over the set of n parties. Then, there exists a secret sharing scheme for Γ whose share size is $O(nk \cdot \text{CDS}(\mathcal{S}_{k,d}) + kn^2 \log n) = nk^3 2^{\tilde{O}(\sqrt{d \log k})} + O(kn^2 \log n)$.*

As a comparison, one can also construct a secret sharing scheme for a d -partite k -uniform access structure Γ , by applying a folklore method inspired by [33] (see, e.g., [22] for a formal description). This construction results in the share size proportional to the total number of minimal vectors in $\mathcal{P}_\Gamma$. Since Γ is d -partite and k -uniform, this number is at most the total number of integer vectors $(x_i)_{i \in [n]}$ such that $x_i \geq 0$ for all i and $\sum_{i \in [d]} x_i \in \{k, k+1\}$. Thus, the share size of this construction is at most $O(k^{d-1})$. Our scheme improves this bound since the share size is asymptotically smaller than $k^{\tilde{O}(\sqrt{d})}$.

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Gen().

1. For each $i \in [n-1]$ and each $a \in \mathbb{Z}_{n+1}$, choose $r_{i,a} \leftarrow \mathbb{Z}_2$.
2. Output $r = (r_{i,a})_{i \in [n-1], a \in \mathbb{Z}_{n+1}}$.

Enc($i, (\mathbf{m}, x_i), r$).

- If $i = 0$, then output $\mu_0 = r_{0,x_0}$.
- If $i \notin \{0, n-1\}$:
 1. For each $a \in \mathbb{Z}_{n+1}$, let $\mu_{i,a} = r_{i,a} - r_{i-1,a-x_i}$.
 2. Output $\mu_i = (\mu_{i,a})_{a \in \mathbb{Z}_{n+1}}$.
- If $i = n-1$:
 1. For each $a \in \mathbb{Z}_{n+1}$, let $r_{n-1,a} = \mathbf{m}[a]$.
 2. For each $a \in \mathbb{Z}_{n+1}$, let $\mu_{n-1,a} = r_{n-1,a} - r_{n-2,a-x_{n-1}}$.
 3. Output $\mu_{n-1} = (\mu_{n-1,a})_{a \in \mathbb{Z}_{n+1}}$.

Dec($\mu_0, \dots, \mu_{n-1}; x_0, \dots, x_{n-1}$).

1. For each $i = 1, \dots, n-1$, let $s_i = \sum_{j=0}^i x_j$.
2. Output $\mu_0 + \sum_{i \in [n]} \mu_{i,s_i}$.

Figure 9: A PSM protocol for pSel_n

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A Proof of Proposition 3.11

We describe a protocol for pSel_n in Figure 9.

The correctness follows from

$$\mu_0 + \sum_{i \in [n]} \mu_{i,s_i} = \sum_{i=0}^{n-1} r_{i,s_i} - \sum_{i=0}^{n-2} r_{i,s_i} = \mathbf{m}[s_{n-1}] = \text{pSel}_n((\mathbf{m}, x_i)_{i \in [n]}),$$

where $s_{n-1} = \sum_{i \in [n]} x_i$ and $s_0 = x_0$.

To see privacy, let $\mathbf{m}, \mathbf{m}' \in \{0,1\}^{n+1}$ and $\mathbf{x} = (x_i)_{i \in [n]}, \mathbf{x}' = (x'_i)_{i \in [n]} \in \{0,1\}^n$ be inputs such that $\text{pSel}_n((\mathbf{m}, x_i)_{i \in [n]}) = \text{pSel}_n((\mathbf{m}', x'_i)_{i \in [n]})$. By the definition of pSel_n , we have that $x'_i = x_i$ for all $i \in [n]$ and $\mathbf{m}[s] = \mathbf{m}'[s]$ where $s = \sum_{i \in [n]} x_i$. We denote by $r = (r_{i,a})_{i \in [n-1], a \in \mathbb{Z}_{n+1}}$ and $r' = (r'_{i,a})_{i \in [n-1], a \in \mathbb{Z}_{n+1}}$ randomness in the execution of Π on inputs $(\mathbf{m}, x_i)_{i \in [n]}$ and $(\mathbf{m}', x'_i)_{i \in [n]}$, respectively. It suffices to show that there is a bijection between r and r' such that messages seen by the referee are identical, i.e., $\text{Enc}(i, (\mathbf{m}, x_i), r) = \text{Enc}(i, (\mathbf{m}', x'_i), r')$ for all $i \in [n]$.

Let $r_{n-1,a} = \mathbf{m}[a]$ and $r'_{n-1,a} = \mathbf{m}'[a]$ for all $a \in \mathbb{Z}_{n+1}$. Define a map from r to r' as

$$r'_{i,a} = r_{i,a} + r'_{i+1,a+x_{i+1}} - r_{i+1,a+x_{i+1}}, \quad \forall i \in [n-1], \forall a \in \mathbb{Z}_{n+1}.$$

Clearly, this map is a bijection since we have an inverse

$$r_{i,a} = r'_{i,a} + r_{i+1,a+x_{i+1}} - r'_{i+1,a+x_{i+1}}, \quad \forall i \in [n-1], \forall a \in \mathbb{Z}_{n+1}.$$

For $2 \leq i \leq n-1$, it holds that

$$\begin{aligned} \text{Enc}(i, (\mathbf{m}', x_i), r') &= r'_{i,a} - r'_{i-1,a-x_i} \\ &= r'_{i,a} - (r_{i-1,a-x_i} + r'_{i,(a-x_i)+x_i} - r_{i,(a-x_i)+x_i}) \\ &= r_{i,a} - r_{i-1,a-x_i} \\ &= \text{Enc}(i, (\mathbf{m}, x_i), r). \end{aligned}$$

It also holds that $r'_{i,s_i} - r_{i,s_i} = r'_{i+1,s_i+x_{i+1}} - r_{i+1,s_i+x_{i+1}} = r'_{i+1,s_{i+1}} - r_{i+1,s_{i+1}}$ for any $i \in [n]$, where $s_i = \sum_{j=0}^i x_j$. Therefore,

$$\text{Enc}(0, (\mathbf{m}', x_0), r') - \text{Enc}(0, (\mathbf{m}, x_0), r) = r'_{0,s_0} - r_{0,s_0} = r'_{n-1,s_{n-1}} - r_{n-1,s_{n-1}} = \mathbf{m}'[s_{n-1}] - \mathbf{m}[s_{n-1}] = 0.$$

The communication complexity of Π is $O(n^2)$ since each party sends at most $n+1$ bits.

B PSM Protocols for Selection Functions

By modifying the PSM protocols in [40, 18], we obtain PSM protocols for $\text{Sel}_{k,n}$. Note that the original constructions for Sel_n can be recovered by setting $k = n$.

Proposition B.1. *For any $k \leq n$, there exist PSM protocols for the partial function $\text{Sel}_{k,n}$ achieving the following efficiency:*

- The communication complexity is $O(kn)$ and the computational complexity is $O(nk^2) + 2^{O(k)}$.
- The communication complexity is $O(kn \log n)$ and the computational complexity is $\tilde{O}(kn)$.

In particular, we have $\text{PSM}(\text{Sel}_{k,n}) = O(kn)$.

The proposition follows from Propositions B.2 and B.3, which are shown in the subsequent sections. Part of the descriptions are taken verbatim from [24].

We note that n -out-of- n ad hoc PSM protocols require an additional privacy property that less than n messages reveal no information on inputs, which is not immediately satisfied by plain PSM protocols. We also prove that the above PSM protocols for $\text{Sel}_{k,n}$ actually satisfy this property and hence they are n -out-of- n ad hoc PSM protocols.

B.1 A Protocol Based on [40]

Let p be a prime. We say that $a \in \mathbb{Z}_p^*$ is a quadratic residue modulo p , denoted by $\left(\frac{a}{p}\right) = 1$, if there exists $r \in \mathbb{Z}_p^*$ such that $r^2 = a \bmod p$. Otherwise, $a \in \mathbb{Z}_p^*$ is said to be a quadratic non-residue, in which case we denote by $\left(\frac{a}{p}\right) = -1$. We denote the sequence of quadratic characters modulo p by $\mathbf{Q}_p \in \{0, 1\}^{p-1}$. I.e., for any $i \in \mathbb{Z}_p^* = \{1, 2, \dots, p-1\}$, $\mathbf{Q}_p[i] = 0$ if $\left(\frac{i}{p}\right) = 1$ and $\mathbf{Q}_p[i] = 1$ if $\left(\frac{i}{p}\right) = -1$.

The ℓ -th Peralta prime p_ℓ [38] is the smallest prime p such that any bit string $\mathbf{T}$ of length ℓ appears in $\mathbf{Q}_p$, i.e., there exists $a \in \mathbb{Z}_p^*$ such that $\mathbf{T}[i] = \mathbf{Q}_p[a+i]$ for all $i \in [\ell]$, or equivalently $\left(\frac{a+i}{p}\right) = (-1)^{\mathbf{T}[i]}$. We call a an *offset* for $\mathbf{T}$ with respect to p_ℓ . It was shown in [38] that for any $\ell \in \mathbb{N}$, there exists the ℓ -th Peralta prime p_ℓ and it holds that $p_\ell = 2^{O(\ell)}$.

Notations. Let p be the k -th Peralta prime.

$\text{Gen}()$.

1. Choose $r_0, \dots, r_{n-1} \in \mathbb{Z}_p$ uniformly at random conditioned on $\sum_{i=1}^n r_i = 0$.
2. Choose $s'_0 \leftarrow \mathbb{Z}_p^*$ and set $s_0 = (s'_0)^2$.
3. Output $r = (s_0, r_0, \dots, s_{n-1})$.

$\text{Enc}(i, (\mathbf{m}, x_i), r)$.

- If $i = 0$:
 1. Find an offset a for $\mathbf{m}$ with respect to p .
 2. Output $\mu_0 = s_0(a + x_0) + r_0$.
- If $i \neq 0$, then output $\mu_i = s_0 x_i + r_i$.

$\text{Dec}(\mu_0, \dots, \mu_{n-1})$.

1. Compute $\bar{\mu} = \sum_{i \in [n]} \mu_i$ and compute $q = \left(\frac{\bar{\mu}}{p}\right) \in \{-1, +1\}$.
2. Output $y = 0$ if $q = 1$ and $y = 1$ if $q = -1$.

Figure 10: A PSM protocol for $\text{Sel}_{k,n}$ based on [40]

Proposition B.2. *The protocol Π described in Figure 10 is a PSM protocol for $\text{Sel}_{k,n}$ with communication complexity $O(kn)$ and computational complexity $O(nk^2) + 2^{O(k)}$.*

Proof. Let $\mathbf{m} \in \{0, 1\}^{k+1}$ and $(x_i)_{i \in [n]} \in \{0, 1\}^n$ be such that $u := \sum_{i \in [n]} x_i \leq k$.

Note that $\bar{\mu} = s_0(a + u)$ and s_0 is a quadratic residue. Since a is an offset for $\mathbf{m}$, it holds that

$$\left(\frac{\bar{\mu}}{p}\right) = \left(\frac{a + u}{p}\right) = (-1)^{\mathbf{m}[u]}.$$

Thus, $y = \mathbf{m}[u]$ and the correctness of Π follows.

Since $(r_i)_{i \in [n]}$ are uniformly random conditioned on $\sum_{i \in [n]} r_i = 0$, the messages $(\mu_i)_{i \in [n]}$ can be simulated from $\bar{\mu} = s_0(a + u)$ only. Since s_0 is uniformly distributed over the set Q of all quadratic residues, $\bar{\mu}$ is uniformly distributed over Q if $\left(\frac{a+u}{p}\right) = 1$, i.e., $\mathbf{m}[u] = 0$; and $\bar{\mu}$ is uniformly distributed over $\mathbb{Z}_p^* \setminus Q$ if $\left(\frac{a+u}{p}\right) = -1$, i.e., $\mathbf{m}[u] = 1$. Thus, the distribution of $(\mu_i)_{i \in [n]}$ depends only on $\mathbf{m}[u]$, which implies the privacy of Π .

The communication complexity of Π is $O(n \log p) = O(kn)$ since $p = 2^{O(k)}$. Party 0 needs to find an offset a for $\mathbf{m}$, which requires an exhaustive search over $\mathbb{Z}_p^*$. This step incurs the computational complexity of $O(p) = 2^{O(k)}$. Since the other parts only involve $O(n)$ arithmetic operations on $\mathbb{Z}_p$, the computational complexity is $O(nk^2)$. Thus, the total computational complexity of Π is $O(nk^2) + 2^{O(k)}$. $\square$

B.2 A Protocol Based on [18, 23]

Proposition B.3. *The protocol Π described in Figure 11 is a PSM protocol for $\text{Sel}_{k,n}$ with communication complexity $O(kn \log n)$ and computational complexity $\tilde{O}(kn)$.*

Proof. Let $\mathbf{m} \in \{0, 1\}^{k+1}$ and $(x_i)_{i \in [n]} \in \{0, 1\}^n$ be such that $u := \sum_{i \in [n]} x_i \leq k$.

First, we see the functionality of the sub-algorithms $(\text{Gen}_0, \text{Enc}_0, \text{Dec}_0)$. Let $\ell \in [k + 1]$. Then, it holds that $\bar{\nu} - \sum_{i \in [n]} \nu_i = t(\ell - u)$ and $\bar{\mu} + \sum_{i \in [n]} \mu_i = \mathbf{m}[\ell] - s(\ell - u)$. Since $t \in \mathbb{Z}_p^*$, it holds that $\bar{\nu} = \sum_{i \in [n]} \nu_i$ if and only if $\ell = u$. In this case, Dec_0 outputs $y = \mathbf{m}[\ell]$.

Next, we see the correctness of the main protocol. Let $\ell \in [k + 1]$. Since $\widetilde{M}_i^{(\ell)} = M_i^{(\pi(\ell))}$, it holds that

$$y^{(\ell)} = \text{Dec}_0((M_i^{(\pi(\ell))})_{i \in [n]}) = \text{Dec}_0((\text{Enc}_0(i, (\mathbf{m}, x_i), r^{(\pi(\ell))}); \pi(\ell)))_{i \in [n]}).$$

Notations. Let p be a prime such that $p > k$.

Sub-protocol. Define algorithms $(\text{Gen}_0, \text{Enc}_0, \text{Dec}_0)$ as follows.

$\text{Gen}_0(\ell)$. On input $\ell \in [k+1]$,

1. Choose $s, t \leftarrow \mathbb{Z}_p^*$.
2. For each $i \in [n]$, choose $u_i, v_i \leftarrow \mathbb{Z}_p$.
3. Output $r = (s, t, (u_i)_{i \in [n]}, (v_i)_{i \in [n]})$.

$\text{Enc}_0(i, (\mathbf{m}, x_i), r; \ell)$. On input $i \in [n]$, $\mathbf{m} \in \{0, 1\}^{k+1}$, $x_i \in \{0, 1\}$, randomness r , and $\ell \in [k+1]$,

- If $i = 0$:
 1. Compute $\bar{\mu} = \mathbf{m}[\ell] - s\ell - \sum_{i \in [n]} u_i$ and $\bar{\nu} = t\ell + \sum_{i \in [n]} v_i$.
 2. Compute $\mu_0 = sx_0 + u_0$ and $\nu_0 = tx_0 + v_0$.
 3. Output $M_0 = (\bar{\mu}, \bar{\nu}, \mu_0, \nu_0)$.
- If $i \neq 0$:
 1. Compute $\mu_i = sx_i + u_i$ and $\nu_i = tx_i + v_i$.
 2. Output $M_i = (\mu_i, \nu_i)$.

$\text{Dec}_0(M_0, \dots, M_{n-1})$. Output

$$y = \begin{cases} \bar{\mu} + \sum_{i \in [n]} \mu_i, & \text{if } \bar{\nu} = \sum_{i \in [n]} \nu_i, \\ \perp, & \text{otherwise.} \end{cases}$$

Main protocol.

$\text{Gen}()$.

1. Choose a uniformly random permutation π over $[k+1]$.
2. For each $\ell \in [k+1]$, run $r^{(\ell)} \leftarrow \text{Gen}_0(\ell)$.
3. Output $r = (\pi, r^{(\ell)})_{\ell \in [k+1]}$.

$\text{Enc}(i, (\mathbf{m}, x_i), r)$.

1. For each $\ell \in [k+1]$, run $M_i^{(\ell)} \leftarrow \text{Enc}_0(i, (\mathbf{m}, x_i), r^{(\ell)}; \ell)$.
2. Output $M_i = (M_i^{(\pi(\ell))})_{\ell \in [k+1]}$.

$\text{Dec}(M_0, \dots, M_{n-1})$.

1. Parse each M_i as $M_i = (\widetilde{M}_i^{(\ell)})_{\ell \in [k+1]}$.
2. For each $\ell \in [k+1]$, run $y^{(\ell)} \leftarrow \text{Dec}_0((\widetilde{M}_i^{(\ell)})_{i \in [n]})$.
3. If there exists a unique $\ell \in [k+1]$ such that $y^{(\ell)} \neq \perp$, then output $y^{(\ell)}$. Otherwise, output $\perp$.

Figure 11: A PSM protocol for $\text{Sel}_{k,n}$ based on [18, 23]

By the functionality of the sub-algorithms, $y^{(\ell)} \neq \perp$ if and only if $\ell = \pi^{-1}(u)$. Therefore, $(y^{(\ell)})_{\ell \in [k+1]}$ contains exactly one non- $\perp$ value and the value is $y^{(\pi^{-1}(u))} = \text{Dec}_0((\text{Enc}_0(i, (\mathbf{m}, x_i), r^{(u)}; u))_{i \in [n]}) = \mathbf{m}[u]$.

We show privacy. For each $\ell \in [k+1]$, we parse $r^{(\ell)}$ as $r^{(\ell)} = (s^{(\ell)}, t^{(\ell)}, (u_i^{(\ell)})_{i \in [n]}, (v_i^{(\ell)})_{i \in [n]})$. We parse each $M_i^{(\ell)}$ as $M_i^{(0)} = (\bar{\mu}^{(\ell)}, \bar{\nu}^{(\ell)}, \mu_0^{(\ell)}, \nu_0^{(\ell)})$ and $M_i^{(\ell)} = (\mu_i^{(\ell)}, \nu_i^{(\ell)})$. That is,

$$\begin{aligned}\bar{\mu}^{(\ell)} &= \mathbf{m}[\ell] - s^{(\ell)}\ell - \sum_{i \in [n]} u_i^{(\ell)}, \quad \bar{\nu}^{(\ell)} = t^{(\ell)}\ell + \sum_{i \in [n]} v_i^{(\ell)}, \\ \mu_i^{(\ell)} &= s^{(\ell)}x_i + u_i^{(\ell)}, \quad \text{and } \nu_i^{(\ell)} = t^{(\ell)}x_i + v_i^{(\ell)}.\end{aligned}$$

The messages that the referee sees are those obtained by randomly permuting $(\bar{\mu}^{(\ell)}, \bar{\nu}^{(\ell)}, (\mu_i^{(\ell)}, \nu_i^{(\ell)})_{i \in [n]})_{\ell \in [k+1]}$ using the permutation π .

Since $(u_i^{(\ell)})_{i \in [n]}$ are independent and uniformly random elements, $(\bar{\mu}^{(\ell)}, (\mu_i^{(\ell)})_{i \in [n]})$ can be viewed as random additive shares of

$$\kappa^{(\ell)} := \bar{\mu}^{(\ell)} + \sum_{i \in [n]} \mu_i^{(\ell)} = \mathbf{m}[\ell] - s^{(\ell)}\ell + s^{(\ell)}u.$$

Thus, $(\bar{\mu}^{(\ell)}, (\mu_i^{(\ell)})_{i \in [n]})$ can be simulated from $\kappa^{(\ell)}$ only. Similarly, since $(v_i^{(\ell)})_{i \in [n]}$ are independent and uniformly random elements, $(\bar{\nu}^{(\ell)}, (\nu_i^{(\ell)})_{i \in [n]})$ can be simulated from

$$\lambda^{(\ell)} := \bar{\nu}^{(\ell)} + \sum_{i \in [n]} \nu_i^{(\ell)} = t^{(\ell)}\ell - t^{(\ell)}u.$$

Therefore, it suffices to show that $(\kappa^{(\pi(\ell))}, \lambda^{(\pi(\ell))})_{\ell \in [k+1]}$ depend only on $\mathbf{m}[u]$. Since each $t^{(\ell)}$ is a uniformly random element in $\mathbb{Z}_p^*$,

$$(\bar{\lambda}^{(0)}, \dots, \bar{\lambda}^{(k)}) = (*, \dots, *, 0, *, \dots, *),$$

where $*$ denotes a uniformly random element in $\mathbb{Z}_p^*$ and the only one 0 is at position u . Thus, the vector $(\lambda^{(\pi(\ell))})_{\ell \in [k+1]}$ consists of one 0 and k uniformly random elements in $\mathbb{Z}_p^*$, with the position of the 0 being uniformly distributed. Furthermore, since each $s^{(\ell)}$ is uniformly distributed over $\mathbb{Z}_p^*$, we have that

$$\begin{aligned}(\bar{\kappa}^{(0)}, \dots, \bar{\kappa}^{(k)}) &= \mathbf{m} - (0 - \bar{\lambda}^{(0)}, \dots, k - \bar{\lambda}^{(k)}) \begin{pmatrix} s^{(0)} & & \\ & \ddots & \\ & & s^{(k)} \end{pmatrix} \\ &= (*, \dots, *, \mathbf{m}[u], *, \dots, *),\end{aligned}$$

where $\mathbf{m}[u]$ is at position u . Thus, the entries of the vector $(\kappa^{(\pi(\ell))})_{\ell \in [k+1]}$ consists of $\mathbf{m}[u]$ and k uniformly random elements in $\mathbb{Z}_p^*$, with the position of $\mathbf{m}[u]$ being uniformly distributed. We therefore conclude that $(\kappa^{(\pi(\ell))}, \lambda^{(\pi(\ell))})_{\ell \in [k+1]}$ depend only on $\mathbf{m}[u]$.

The communication complexity of Π is $O(nk \log k)$ since each party's message consists of $O(k)$ elements of $\mathbb{Z}_p$ and we can choose a prime p such that $p < 2k$. Since all the steps consist of at most $O(kn)$ arithmetic operations on $\mathbb{Z}_p$, the computational complexity is $\tilde{O}(kn)$. $\square$

Additionally, these protocols ensure that no information is leaked on inputs when the referee receive less than n messages, since the messages are masked with random additive shares of zero.

C Proof of Theorem 6.2

Before proceeding to the proof, we recall the (k, n) -Shamir secret sharing [39], which is the most common secret sharing scheme for a threshold access structure $\Gamma = \{A : |A| \geq k\}$. Let $\mathbb{F}_q$ be a finite field of size $q > n$. The i^{th} share is set as $v_i = p(i)$ for a random degree- $(k-1)$ polynomial p over $\mathbb{F}_q$ such that

Notations.

- Let $\Pi_{f^*} = (\text{Gen}_{f^*}, \text{Enc}_{f^*}, \text{Dec}_{f^*})$ be a $(d-1)$ -party PSM protocol for f^* .
- Let $\Pi_{\text{sel}} = (\text{Gen}_{\text{sel}}, \text{Enc}_{\text{sel}}, \text{Dec}_{\text{sel}})$ be an n -out-of- n ad hoc PSM protocol for the partial function $\text{Sel}_{k,n}$.
- Let P_i denote the i^{th} input party.

Randomness generation.

1. Run $r_{f^*} \leftarrow \text{Gen}_{f^*}()$.
2. For each $j \in [d-1]$, run $r_{\text{sel}}^{(j)} \leftarrow \text{Gen}_{\text{sel}}()$ with independent randomness.
3. For each $j \in [d-1]$ and each $\ell \in [k+1]$, compute a message $s_\ell^{(j)} = \text{Enc}_{f^*}(j, \ell, r_{f^*})$ of the j^{th} party in Π_{f^*} encoding an input $\ell \in [k+1]$. Let $\mathbf{s}^{(j)} = (s_\ell^{(j)})_{\ell \in [k+1]}$.
4. Give $((\mathbf{s}^{(j)})_{j \in [d-1]}, (r_{\text{sel}}^{(j)})_{j \in [d-1]})$ to every party P_i .
5. Additionally, for each $j \in [d-1]$ and each $h \in [n]$, compute a message $t_h^{(j)} = \text{Enc}_{\text{sel}}(h, (\mathbf{s}^{(j)}, 0), r_{\text{sel}}^{(j)})$ of the h^{th} party in Π_{sel} encoding an input $(\mathbf{s}^{(j)}, 0)$.
6. For each $j \in [d-1]$ and each $h \in [n]$, distribute k -out-of- $(n-1)$ secret shares of $t_h^{(j)}$ among parties except for P_h . Let $\llbracket t_h^{(j)} \rrbracket_i$ denote the share that P_i receives.

Sending messages. P_i with input $x_i \in [d]$ does the following.

1. Let $\phi(x_i) = (v_i^{(0)}, \dots, v_i^{(d-2)}) \in \{0, 1\}^{d-1}$.
2. For each $j \in [d-1]$, compute $m_i^{(j)} = \text{Enc}_{\text{sel}}(i, (\mathbf{s}^{(j)}, v_i^{(j)}), r_{\text{sel}}^{(j)})$.
3. Send the referee $(m_i^{(j)})_{j \in [d-1]}$ and $(\llbracket t_h^{(j)} \rrbracket_i)_{j \in [d-1], h \in [n] \setminus \{i\}}$.

Output. Upon receiving messages from a set A of k parties, the referee does the following.

1. For each $j \in [d-1]$ and $h \in [n] \setminus A$, reconstruct $t_h^{(j)}$ from the k shares $(\llbracket t_h^{(j)} \rrbracket_i)_{i \in A}$.
2. For each $j \in [d-1]$, compute $u^{(j)} = \text{Dec}_{\text{sel}}((m_i^{(j)})_{i \in A}, (t_h^{(j)})_{h \in [n] \setminus A})$.
3. Output $y = \text{Dec}_{f^*}(u^{(0)}, \dots, u^{(d-2)})$.

Figure 12: An ad hoc PSM protocol for $\mathcal{S}_{k,d}$

$p(0) = s$. Note that the (k, n) -Shamir scheme can share any field element $s \in \mathbb{F}_q$, i.e., a bit string of length $\ell_0 = \lceil \log q \rceil = O(\log n)$, as a secret without increasing the share size. To share a bit string $s \in \{0, 1\}^\ell$, we can divide s into $\lceil \ell / \ell_0 \rceil$ sub-strings of length ℓ_0 each. Applying the (k, n) -Shamir scheme to such strings results in each share of size $\lceil \ell / \ell_0 \rceil \cdot \ell_0 \leq \ell + \ell_0 = \ell + O(\log n)$. We denote the i^{th} share of a secret s under the (k, n) -Shamir scheme by $\llbracket s \rrbracket_i$.

Let $f^* : [k+1]^{d-1} \rightarrow \{0, 1\}$ be a function associated with f satisfying Eq. (8). Let $\Pi_{f^*} = (\text{Gen}_{f^*}, \text{Enc}_{f^*}, \text{Dec}_{f^*})$ be a $(d-1)$ -party PSM protocol for f^* . Let $\Pi_{\text{sel}} = (\text{Gen}_{\text{sel}}, \text{Enc}_{\text{sel}}, \text{Dec}_{\text{sel}})$ be an n -out-of- n ad hoc PSM protocol for the partial function $\text{Sel}_{k,n}$. We describe a k -out-of- n ad hoc PSM protocol Π for f in Figure 12.

Correctness. Let $A \subseteq [n]$ denote a subset of k parties who actually participate in the protocol. Without loss of generality, we may assume that $A = [k] = \{0, 1, \dots, k-1\}$. Let $(x_i)_{i \in [k]} \in [d]^k$ be the inputs of parties in $A = [k]$. For $i \in [k]$, let $\phi(x_i) = (v_i^{(j)})_{j \in [d-1]} \in \{0, 1\}^{d-1}$. We define $v_i^{(j)} = 0$ for $i \in [n] \setminus [k]$ and $j \in [d-1]$. We denote $\ell^{(j)} = \sum_{i \in [n]} v_i^{(j)}$.

For each $h \in [n] \setminus [k]$, $t_h^{(j)}$ can be reconstructed from k shares $\llbracket t_h^{(j)} \rrbracket_0, \dots, \llbracket t_h^{(j)} \rrbracket_{k-1}$ held by A . We have

that for each $h \in [n] \setminus [k]$ and $j \in [d-1]$,

$$t_h^{(j)} = \text{Enc}_{\text{sel}}(h, (\mathbf{s}^{(j)}, 0), r_{\text{sel}}^{(j)}) = \text{Enc}_{\text{sel}}(h, (\mathbf{s}^{(j)}, v_h^{(j)}), r_{\text{sel}}^{(j)}).$$

We also have that for each $i \in [k]$ and $j \in [d-1]$,

$$m_i^{(j)} = \text{Enc}_{\text{sel}}(i, (\mathbf{s}^{(j)}, v_i^{(j)}), r_{\text{sel}}^{(j)}).$$

Thus, since $\sum_{i \in [n]} v_i^{(j)} \leq k$, it holds that for each $j \in [d-1]$,

$$u^{(j)} = \text{Dec}_{\text{sel}}(m_0^{(j)}, \dots, m_{k-1}^{(j)}, t_k^{(j)}, \dots, t_{n-1}^{(j)}) = \text{Sel}_{k,n}((\mathbf{s}^{(j)}, v_i^{(j)})_{i \in [n]}) = s_{\ell^{(j)}}^{(j)} = \text{Enc}_{f^*}(j, \ell^{(j)}, r_{f^*}).$$

and hence that

$$y = \text{Dec}_{f^*}(u^{(0)}, \dots, u^{(d-2)}) = f^*(\ell^{(0)}, \dots, \ell^{(d-2)}) = f(x_0, \dots, x_{n-1}).$$

Privacy. Let A be a set of parties who actually send messages to the referee. Then, the referee learns the following elements:

$$(m_i^{(j)})_{j \in [d-1]}, (\llbracket t_h^{(j)} \rrbracket_i)_{j \in [d-1], h \in [n] \setminus \{i\}}, \forall i \in A.$$

First, suppose that $|A| = k' < k$. Then, for any $h \in [n]$, $(\llbracket t_h^{(j)} \rrbracket_i)_{i \in A \setminus \{h\}}$ reveal no information on $t_h^{(j)}$, as the number of shares is less than k . Furthermore, $(m_i^{(j)})_{i \in A}$ can be simulated from A only, since Π_{sel} is an n -out-of- n ad hoc PSM protocol. Therefore, the second privacy requirement in Definition 6.1 is satisfied.

Next, suppose that $|A| = k$. For any $h \in A$, $(\llbracket t_h^{(j)} \rrbracket_i)_{i \in A \setminus \{h\}}$ reveal no information on $t_h^{(j)}$, as the number of shares is at most $|A \setminus \{h\}| = k-1$. The remaining pieces to simulate are

$$(m_i^{(j)})_{i \in A}, (\llbracket t_h^{(j)} \rrbracket_i)_{i \in A, h \in \bar{A}} = (\text{Enc}_{\text{sel}}(i, (\mathbf{s}^{(j)}, v_i^{(j)}), r_{\text{sel}}^{(j)}))_{i \in A}, (\llbracket \text{Enc}_{\text{sel}}(h, (\mathbf{s}^{(j)}, 0), r_{\text{sel}}^{(j)}) \rrbracket_A)_{h \in \bar{A}} \quad (10)$$

for all $j \in [d-1]$, where we denote $\llbracket x \rrbracket_A = (\llbracket x \rrbracket_i)_{i \in A}$. For each $h \in \bar{A}$, the shares $\llbracket \text{Enc}_{\text{sel}}(h, (\mathbf{s}^{(j)}, 0), r_{\text{sel}}^{(j)}) \rrbracket_A$ are simulated from $\text{Enc}_{\text{sel}}(h, (\mathbf{s}^{(j)}, 0), r_{\text{sel}}^{(j)})$ by executing the sharing algorithm on it. Note that

$$(\text{Enc}_{\text{sel}}(i, (\mathbf{s}^{(j)}, v_i^{(j)}), r_{\text{sel}}^{(j)}))_{i \in A}, (\text{Enc}_{\text{sel}}(h, (\mathbf{s}^{(j)}, 0), r_{\text{sel}}^{(j)}))_{h \in \bar{A}}$$

are simulated from $s_{\ell^{(j)}}^{(j)}$ by the simulator of Π_{sel} , where $\ell^{(j)} = \sum_{i \in A} v_i^{(j)}$. Furthermore, since $s_{\ell^{(j)}}^{(j)} = \text{Enc}_{f^*}(j, \ell^{(j)}, r_{f^*})$, we can simulate $(s_{\ell^{(j)}}^{(j)})_{j \in [d-1]}$ from $f^*(\ell^{(0)}, \dots, \ell^{(d-2)}) = f(x_0, \dots, x_{n-1})$ by using the simulator of Π_g . Therefore, we conclude that the elements (10) are simulated from $f(x_0, \dots, x_{n-1})$ only. This implies that the first privacy requirement in Definition 6.1 is satisfied.

Complexity. When a set $A = [k]$ of parties actually participate in the protocol, the referee receives the following elements:

$$(m_i^{(j)})_{j \in [d-1]}, (\llbracket t_h^{(j)} \rrbracket_i)_{j \in [d-1], h \in [n] \setminus \{i\}}, \forall i \in [k].$$

We have that $|s_{\ell^{(j)}}^{(j)}| \leq \text{comm}_j(\Pi_{f^*})$, where $\text{comm}_j(\Pi_{f^*})$ denote the bit length of messages of the j^{th} party in Π_{f^*} . We thus have that

$$|(m_0^{(j)}, \dots, m_{k-1}^{(j)})| \leq \text{Comm}(\Pi_{\text{sel}}) \cdot \text{comm}_j(\Pi_{f^*}) =: N^{(j)}$$

and that $\sum_{i \in [k], j \in [d-1]} |m_i^{(j)}| \leq \sum_{j \in [d-1]} N^{(j)} \leq \text{Comm}(\Pi_{\text{sel}}) \cdot \text{PSM}(f^*)$. Furthermore, we have that

$$|(t_0^{(j)}, \dots, t_{n-1}^{(j)})| \leq \text{Comm}(\Pi_{\text{sel}}) \cdot \text{comm}_j(\Pi_{f^*}) = N^{(j)}.$$

Notations.

- Let $h_\Gamma \in \mathcal{S}_{k,d}$ be the function defined in (9).
- Let $\Pi_h = (\text{Gen}_h, \text{Enc}_h, \text{Dec}_h)$ be a CDS protocol for h_Γ .

The sharing algorithm. Given a secret $s \in \{0, 1\}$,

1. Share s using the $(k+1, n)$ -Shamir secret sharing scheme among the n parties. That is, give the i^{th} share $\llbracket s \rrbracket_i$ to P_i .
2. Run $r \leftarrow \text{Gen}_h()$.
3. For each $j \in [k]$ and $m \in [d]$, compute $\mu_j(m) = \text{Enc}_h(j, m, s, r) \in \{0, 1\}^{\text{comm}_j}$.
4. For each $j \in [k]$ and $i \in [n]$, choose $r_{j,i} \leftarrow \{0, 1\}^{\text{comm}_i}$ if $j \neq 0$ and set $r_{0,i} = 0$ (i.e., a string of comm_i zeros).
5. For each $j \in [k]$ and $i \in [n]$, choose $q_{j,i} \leftarrow \{0, 1\}^{\text{comm}_i}$ if $j \neq k-1$ and set $q_{k-1,i} = 0$ (i.e., a string of comm_i zeros).
6. For each $i \in [n]$, give $\{\mu_j(m_i) \oplus r_{j,i} \oplus q_{j,i} : j \in [k]\}$ to P_i , where $\oplus$ denotes the bit-wise XOR and m_i is the unique index $m \in [d]$ such that $\bar{n}_m \leq i < \bar{n}_{m+1}$.
7. For each $j \in [k] \setminus \{0\}$ and $i \in [n]$, share $r_{j,i}$ using the $(j, i-1)$ -Shamir secret sharing scheme among $\{P_0, \dots, P_{i-1}\}$.
8. For each $j \in [k] \setminus \{k-1\}$ and $i \in [n]$, share $q_{j,i}$ using the $(k-j-1, n-i-1)$ -Shamir secret sharing scheme among $\{P_{i+1}, \dots, P_{n-1}\}$.

Figure 13: A secret sharing scheme for Γ

Then, we have that

$$\left| \begin{pmatrix} \llbracket t_0^{(j)} \rrbracket_0 & \cdots & \llbracket t_{n-1}^{(j)} \rrbracket_0 \\ \vdots & & \vdots \\ \llbracket t_0^{(j)} \rrbracket_{k-1} & \cdots & \llbracket t_{n-1}^{(j)} \rrbracket_{k-1} \end{pmatrix} \right| \leq k(N^{(j)} + O(\log n))$$

and hence that

$$\sum_{i \in [k], j \in [d-1]} \sum_{h \in [n] \setminus \{i\}} |\llbracket t_h^{(j)} \rrbracket_i| \leq \sum_{j \in [d-1]} k(N^{(j)} + O(\log n)) \leq k \cdot \text{Comm}(\Pi_{\text{sel}}) \cdot \text{PSM}(f^*) + O(kd \log n).$$

From Proposition B.1, we can instantiate Π_{sel} with the one with $\text{Comm}(\Pi_{\text{sel}}) = O(kn)$. Therefore, we conclude that the communication complexity of Π is

$$\text{Comm}(\Pi_{\text{sel}}) \cdot \text{PSM}(f^*) + k \cdot \text{Comm}(\Pi_{\text{sel}}) \cdot \text{PSM}(f^*) + O(kd \log n) \leq O(k^2 n) \cdot \text{PSM}(\mathcal{F}_{d-1, k+1}) + O(kd \log n).$$

D Proof of Theorem 6.4

Let $\{P_0, \dots, P_n\}$ denote the set of n parties. Assume that Γ is d -partite with respect to the partition $(S_m)_{m \in [d]}$. Let $n_m = |S_m|$ and define $\bar{n}_m = \sum_{m' \in [m]} n_{m'}$ if $m > 0$ and $\bar{n}_0 = 0$. Without loss of generality, we may assume that $S_m = \{P_{\bar{n}_m}, \dots, P_{\bar{n}_{m+1}-1}\}$. We also denote $Q_{<i} = \{P_0, \dots, P_{i-1}\}$ and $Q_{>i} = \{P_{i+1}, \dots, P_n\}$. We show a secret sharing scheme in Figure 13.

Correctness. Let $A \in \Gamma$. We show that the parties in A can reconstruct the secret s from their shares. This is clear if $|A| \geq k+1$ since s is shared using the $(k+1, n)$ -Shamir secret sharing scheme.

Suppose that $|A| = k$ and write $A = \{P_{i_0}, \dots, P_{i_{k-1}}\}$, where $i_0 < \dots < i_{k-1}$. For each $j \in [k]$, let m_j be the unique index $m \in [d]$ such that $\bar{n}_m \leq i_j < \bar{n}_{m+1}$, that is, $P_{i_j} \in S_m$. By the definition of h_Γ ,

it holds that $h_\Gamma(m_0, \dots, m_{d-1}) = 1$. For each $j \in \{1, 2, \dots, k\}$, since $|A \cap Q_{<i_j}| = j$ and r_{j,i_j} is shared using the $(j, i_j - 1)$ -Shamir scheme among $Q_{<i_j}$, the parties in A can reconstruct r_{j,i_j} . Similarly, for each $j \in \{0, 1, \dots, k-2\}$, since $|A \cap Q_{>i_j}| = k - j - 1$ and q_{j,i_j} is shared using the $(k - j - 1, n - i_j - 1)$ -Shamir scheme among $Q_{>i_j}$, they can also reconstruct q_{j,i_j} . Furthermore, each P_{i_j} has $\mu_j(m_j) \oplus r_{j,i_j} \oplus q_{j,i_j}$. Thus, the parties in A obtain $\mu_j(m_j)$ for all $j \in [k]$. Since $h_\Gamma(m_0, \dots, m_{k-1}) = 1$, the correctness of Π_h ensures that s is recovered from $(\mu_j(m_j))_{j \in [k]}$.

Privacy. Let $A \notin \Gamma$. We show that the parties in A learn no information on the secret s .

First, suppose that $|A| = k$ and write $A = \{P_{i_0}, \dots, P_{i_{k-1}}\}$, where $i_0 < \dots < i_{k-1}$. For each $j \in [k]$, let m_j be the unique index $m \in [d]$ such that $P_{i_j} \in S_m$. As in the proof of correctness, they learn $\mu_j(m_j)$ for all $j \in [k]$. Since $h_\Gamma(m_0, \dots, m_{k-1}) = 0$, the privacy of the CDS protocol Π_h ensures that these messages reveal no information on s . It suffices to show that they cannot learn any other message of Π_h . Clearly, they cannot learn $\mu_j(m)$ for any $j \in [k]$ and $m \in [d] \setminus \{m_0, \dots, m_{k-1}\}$ since a message related to $\mu_j(m)$ is received only by parties in S_m , which is disjoint from A in this case. It remains to show that the parties in A cannot learn $\mu_j(m)$ for any $j \in [k]$ and $m \in \{m_0, \dots, m_{k-1}\} \setminus \{m_j\}$. The messages related to $\mu_j(m)$ that they can learn are $\{\mu_j(m) + r_{j,i} + q_{j,i} : P_i \in A \cap S_m\}$. First, suppose that $m < m_j$. Then, the parties in A cannot recover $r_{j,i}$ for any i with $P_i \in A \cap S_m$. Indeed, letting $j' \in [k]$ be such that $i = i_{j'}$, we have that $|A \cap Q_{<i_{j'}}| = |\{P_{i_0}, \dots, P_{i_{j'-1}}\}| = j' \leq j - 1$. Since $r_{j,i}$ is shared among $Q_{<i}$ using $(j, i - 1)$ -Shamir secret sharing scheme, the parties in A cannot recover it. Next, suppose that $m > m_j$. Then, they cannot recover $q_{j,i}$ for any i with $P_i \in A \cap S_m$. Letting $j' \in [k]$ be such that $i = i_{j'}$, we have that $|A \cap Q_{>i_{j'}}| = |\{P_{i_{j'+1}}, \dots, P_{i_{k-1}}\}| = k - j' - 1 \leq k - j - 2$. Since $q_{j,i}$ is shared among $Q_{>i}$ using $(k - j - 1, n - i - 1)$ -Shamir secret sharing scheme, the parties in A cannot recover $q_{j,i}$. Therefore, they cannot learn $\mu_j(m)$ for any $j \in [k]$ and $m \in [d] \setminus \{m_j\}$.

Next, suppose that $|A| < k$. We may assume that $|A| = k - 1$ and write $A = \{P_{i_0}, \dots, P_{i_{k-2}}\}$, where $i_0 < \dots < i_{k-2}$. Let m_j be such that $P_{i_j} \in S_{m_j}$. Similarly to the case of $|A| = k$, the parties in A cannot learn a message $\mu_j(m)$ for any $j \in [k]$ and $m \in [d] \setminus \{m_j\}$. Furthermore, in this case, they cannot learn $\mu_j(m_j)$ for any $j \in [k]$. Indeed, the messages related to $\mu_j(m_j)$ that they obtain are received only by $A \cap S_{m_j}$. Let P_i be any party in $A \cap S_{m_j}$ and let j' be such that $i = i_{j'}$. If $j' < j$, then $|A \cap Q_{<i_{j'}}| \leq j' < j$ and the parties in A cannot recover $r_{j,i}$. If $j' \geq j$, then $|A \cap Q_{>i_{j'}}| \leq |\{P_{i_{j'+1}}, \dots, P_{i_{k-2}}\}| \leq k - j - 2 < k - j - 1$ and they cannot recover $q_{j,i}$. Therefore, they cannot recover $\mu_j(m_j)$. The privacy is satisfied as any message of Π_h is revealed.

Complexity. The size of shares distributed at Step 1 is $O(n \log n)$. The size of masked messages distributed at Step 6 is $\sum_{i \in [n]} k \cdot \text{comm}_i = k \cdot \text{Comm}(\Pi_g)$. Since $|r_{j,i}| = \text{comm}_i$, the size of shares distributed at Step 7 is

$$\sum_{j \in [k] \setminus \{0\}} \sum_{i \in [n]} \sum_{\ell=0}^{i-1} |\llbracket r_{j,i} \rrbracket_\ell| \leq k \sum_{i \in [n]} O(n \cdot \text{comm}_i + n \log n) = O(kn \cdot \text{Comm}(\Pi_h) + kn^2 \log n)$$

Similarly, the size of shares distributed at Step 8 is at most $O(kn \cdot \text{Comm}(\Pi_h) + kn^2 \log n)$. Therefore, since $h_\Gamma \in \mathcal{S}_{k,d}$, the total share size is $O(kn \cdot \text{CDS}(\mathcal{S}_{k,d}) + kn^2 \log n)$.